

Decoupled multi-observer design for disturbance estimation with low-order internal models

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ABSTRACT

Implementable applications are often compromised by signals that allow some algebraic representation. Those signals, namely disturbances, negatively affect the performance of control strategies, especially in high-performance contexts. Common issues with existing methods that try to estimate and compensate for those disturbances include complex dimensions, inflexibility, cumbersome design processes, and problems related to high-gain observers that effectively compensate for the disturbances considered in the scheme. This paper introduces a novel, structurally simple multi-observer scheme that leverages prior knowledge of disturbances. By incorporating internal models for each disturbance component, this strategy enhances design simplicity, increases flexibility by allowing the addition of new observers for emerging disturbances, and circumvents problems associated with high dimensions and high gain. Additionally, a novel solution is presented to address the estimation of periodic signals without including a high-dimension observer. The effectiveness of this approach is demonstrated through well-executed and convincing experimental results.

1. Introduction

Despite significant advances in robust control, disturbance rejection remains a central research topic. This is mainly due to the ubiquity of disturbances in practical systems and to their often complex, time-varying structure, which makes accurate modelling and compensation challenging. Classical solutions typically rely on disturbance observers and/or explicit internal models embedded in the controller.

Among disturbance-observer-based strategies, Disturbance Observer-Based Control (DOBC) offers a simple and widely used framework: a linear nominal plant model is complemented with a low-pass filter to estimate input-coupled disturbances, which are then compensated in the control law; see, for example, Babayomi et al. (2024), Li et al. (2024a), Park and Back (2024), Suleimenov and Do (2023), Viswambharan (2025). While effective for low-frequency perturbations, DOBC schemes often lack explicit internal models of structured disturbances and may become conservative when multiple disturbance components are present.

Active Disturbance Rejection Control (ADRC), initially proposed by Han (2009), estimates the so-called *total disturbance* through an Extended State Observer (ESO). Sensitivity analyses, e.g. Sira-Ramírez et al. (2021), show that ADRC achieves good rejection for low-frequency input-coupled disturbances. ADRC has been successfully applied in power electronics and power generation (Chahir et al., 2025; Latif et al., 2025; Zeng et al., 2025), as well as in mechanical and marine systems (Gao et al., 2025; Li et al., 2024b). Recent works explore the use of multiple ESOs to improve performance (Ochani et al., 2025; Tang et al., 2024), reinforcing the interest in multi-observer architectures.

The Internal Model Principle (IMP) provides a complementary route: by embedding a model of the disturbance (or reference) in the controller, exact rejection or tracking is achieved when the internal model matches the signal structure. IMP-based designs have been used, for instance, for periodic biomedical signals (Dey et al., 2022) and for rejecting periodic disturbances in motion and process control (Gan et al., 2025; Vergara-Ramírez et al., 2025; Zhao et al., 2025). Related internal-model control (IMC) extends the use of IMP to state machine architectures, neuronal

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networks, and time-delay systems (Kumar et al., 2024; Meng et al., 2024; Srikanth & Pavan Kumar, 2024). However, incorporating several internal models into a single controller or observer tends to increase the order of the design and to reduce structural flexibility.

A natural way to combine ADRC and IMP is to embed disturbance internal models inside observers, so that the disturbance is first estimated and then compensated. This methodology has been studied from a theoretical standpoint in Joo et al. (2016), Trumpf et al. (2014) and has been applied in power electronics, drives and electric vehicles in Sariyildiz (2025), Sun et al. (2024), Wang et al. (2024). These approaches exploit accurate disturbance models but typically aggregate all internal models into a single high-order observer.

In summary, existing schemes usually face one or more of the following limitations:

- **Suboptimal disturbance rejection:** Single ESOs or DOBC structures without explicit internal models may provide imprecise estimates for time-varying or structured disturbances.
- **Structural rigidity:** Embedding internal models directly in the controller or in a single observer makes it difficult to incorporate or remove disturbance components without redesign.
- **Limited handling of multiple disturbances:** Adding new disturbance components generally requires modifying the entire controller/observer structure and retuning the design.
- **High-dimensional designs:** A single observer that includes multiple internal models leads to large matrices and numerically demanding gain computations.
- **High gain issues:** High-order observers are prone to large gains, peaking phenomena, and sensitivity to measurement noise.

This paper introduces a disturbance-observer-based control scheme aimed at mitigating these drawbacks. The proposed approach employs a *modular multi-observer architecture* in which each disturbance component is associated with its own low-order observer and internal model. The observers are uncoupled in their dynamics but coordinated through the control law, enabling the inclusion or exclusion of disturbance components without redesigning the entire structure. In addition, a novel discrete-time observer is developed for periodic disturbances, based on the internal model of a generic periodic signal, and seamlessly integrated into the same framework. Overall, the scheme achieves disturbance rejection through low-order, low-gain observers, alleviating the numerical and tuning difficulties of large-scale ESOs.

The technique can also be applied to nonlinear differentially flat systems (Sira-Ramírez & Agrawal, New York, 2004). For such systems, a scalar flat output parameterises the states and input, and the dynamics can be rewritten as a linear discrete-time model with an input-coupled *total disturbance* that collects the nonlinear terms. When this representation is available, the proposed observer framework applies directly to the transformed system, so that the nonlinear case is covered via a flatness-based or feedback-linearisation step.

The main contributions of this paper are:

- A novel disturbance observer-based control scheme with a modular architecture, using uncoupled multi-observer structures, each associated with a specific internal model and coordinated control laws. This enables the flexible inclusion or exclusion of disturbance components without redesign.
- An active disturbance rejection strategy that improves complexity and reduces structural rigidity by relying on low-order observers in a low-gain framework, thus avoiding the challenges of large-scale ESOs while preserving high disturbance-rejection performance.
- A new discrete-time observer for periodic disturbances, seamlessly integrated into the multi-observer structure, which enhances the estimation and rejection of periodic signals.

These novelties lead to the following highlight features:

- Flexibility to incorporate different internal models into an existing scheme without redesign, enabling active and simultaneous rejection

of several disturbance types and online activation/deactivation of selected observers.

- An error-based design that induces robust tracking of the signals represented by the internal models.
- The proposed disturbance-observer framework can be combined with feedback linearization for nonlinear systems that admit such a representation, as outlined in Section 2. In this paper, we focus on the linear discrete-time model Eq.(3), while the nonlinear case is presented as a direct extension.

The structure of the paper is as follows. Section 2 presents the problem formulation. Section 3 reviews the classical Extended State Observer with multiple internal models, which is used as a benchmark. Section 4 details the proposed decoupled observers scheme, organised in design stages. Section 5 introduces the extension to periodic-signal observers within the same framework. Section 6 describes the methodology for designing and implementing the decoupled observers. Section 7 validates the approach on an electromechanical system through experimental results. Finally, Section 8 concludes the paper and outlines future research directions.

2. Problem formulation

This section addresses the design challenges inherent in the classical approach to disturbance estimation and attenuation, which traditionally relies on a single extended state observer (ESO) with multiple internal models, as outlined below.

Consider first a nonlinear discrete-time system of the form

$$\begin{aligned} z(k+1) &= \varphi(z(k), u(k)), & z \in \mathbb{R}^n, u \in \mathbb{R}, \\ w(k) &= h(z(k)), & w \in \mathbb{R}, \end{aligned} \quad (1)$$

where $z(k)$ is the state, $u(k)$ is the input and $w(k)$ is the measured output.

If the system is differentially flat and the output w is chosen as a flat output (see, e.g., Sira-Ramírez and Agrawal (New York, 2004)), then it can be written in the following compact form:

$$w(k+n) = (h \circ \varphi^n)(z(k), u(k)), \quad (2)$$

where, for instance,

$$\begin{aligned} w(k+1) &= h(z(k+1)) = h(\varphi(z(k), u(k))) \\ &= (h \circ \varphi)(z(k), u(k)), \end{aligned}$$

and the recursive composition operator is defined as

$$(h \circ \varphi^r) = (h \circ \varphi^{r-1}) \circ \varphi, \quad (h \circ \varphi^0) = h.$$

The flatness condition for the representation in Eq.(2) implies that

$$\frac{\partial w(k+j)}{\partial u(k)} = 0, \quad j = 0, 1, \dots, n-1,$$

$$\frac{\partial w(k+n)}{\partial u(k)} \neq 0.$$

Therefore, by defining

$$\begin{bmatrix} w(k+1) \\ \vdots \\ w(k+n-1) \\ w(k+n) \end{bmatrix} = \begin{bmatrix} x_1(k+1) \\ \vdots \\ x_{n-1}(k+1) \\ x_n(k+1) \end{bmatrix},$$

there exists a diffeomorphism between the original nonlinear discrete-time system Eq.(1) and the following perturbed linear system in control-

table canonical form:

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ \vdots \\ x_{n-1}(k+1) \\ x_n(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ \vdots \\ x_{n-1}(k) \\ x_n(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \frac{\partial w(k+n)}{\partial u(k)} \end{bmatrix} (u(k) + \xi(k)),$$

or, in compact form,

$$\begin{aligned} \mathbf{x}_p(k+1) &= A_p \mathbf{x}_p(k) + B_p (u(k) + \xi(k)), \\ y(k) &= C_p \mathbf{x}_p(k) = x_1(k), \end{aligned} \quad (3)$$

where $\mathbf{x}_p(k) = [x_1(k) \ x_2(k) \ \dots \ x_n(k)]^\top$ and

$$\xi(k) = \frac{w(k+n) - \frac{\partial w(k+n)}{\partial u(k)} u(k)}{\frac{\partial w(k+n)}{\partial u(k)}}.$$

This transformation provides the structure required to apply the results developed in this work to nonlinear differentially flat systems.

Remark 1 (Scope of the nonlinear case). The nonlinear formulation in Eqs.(1)-(3) is not a standing assumption for all the results in this paper. It is presented as a constructive way to recover the disturbance-augmented linear model Eq.(3), when a flatness-based representation is available. In practice, the proposed observer-controller structure applies to any plant for which a discrete-time model of this form can be obtained, for instance via identification or local linearization around an operating point. In particular, when a nonlinear plant is feedback linearized or flatness-based techniques are used to obtain an input-output linear representation, the resulting dynamics can be written in the disturbance-augmented form Eq.(3), so that the observer-controller structures developed in the following sections apply without modification. All simulation and experimental results reported in this paper are, however, based on linear or identified models of the form Eq.(3); the nonlinear case is included to highlight the potential extension of the framework rather than to introduce an additional design step in the examples.

For linear systems, the basic design model is directly given by the set of matrices $\{A_p, B_p, C_p\}$ available from the outset, for example, obtained from a previous identification procedure.

In line with the ADRC philosophy, the term $\xi(k)$ aggregates all additional nonlinear or uncertain behaviours not captured by the base model $\{A_p, B_p, C_p\}$. This additive disturbance term accounts for: (1) endogenous uncertainties associated with parameter variations and modelling errors in the approximate base model used for design; (2) the effect of unmodelled dynamics, representing either unknown or intentionally neglected state-dependent terms of linear or nonlinear nature; and (3) exogenous disturbances such as unknown, unstructured disturbance signals and errors related to approximate discretisation.

The simplified model $\{A_p, B_p, C_p\}$ can be selected from different perspectives depending on the available knowledge of the plant dynamics. For instance, it may be derived from a linearisation around a given operating point or, in a more approximate fashion, chosen as a discrete-time chain of integrators following the ADRC approach (Sira-Ramírez & Agrawal, New York, 2004).

In this work, the disturbance signal $\xi(k)$ in the control loop is treated under the following assumptions:

- The total disturbance signal $\xi(k)$ is decomposed into several components $\xi_i(k)$, each one associated with a specific internal model

characterised by a polynomial difference equation, plus a residual disturbance component:

$$\xi(k) = \xi_1(k) + \xi_2(k) + \dots + \xi_r(k) + \xi_{\text{res}}(k), \quad (4)$$

where $\xi_{\text{res}}(k)$ denotes the residual disturbance.

- If the disturbance includes periodic components, the corresponding periods are known.

Remark 2 (Internal-model based disturbance rejection). The decomposition in Eq.(4) and the assumption that each component $\xi_i(k)$ is generated by a polynomial difference equation are consistent with the internal model principle for disturbance rejection. Embedding such internal models into the observer-controller structure enables, in the nominal case, asymptotic rejection of polynomial and periodic disturbances. The proposed decoupled observer architecture preserves these internal-model properties while improving numerical conditioning and design flexibility with respect to a single ESO that aggregates all the internal models into one extended state.

The above setting leads to two main design objectives, which are summarised next.

2.1. Formal problem statement

We summarise the previous discussion into the following assumption and design problems.

Assumption 1 (Plant and disturbance structure). The plant is described by the discrete-time SISO model Eq.(3), and the total disturbance signal is decomposed as in Eq.(4). Each structured component $\xi_i(k)$, $i = 1, \dots, r$, is generated by a finite-dimensional internal model of the form

$$\mathbf{x}_{\xi_i}(k+1) = A_{\xi_i} \mathbf{x}_{\xi_i}(k), \quad \xi_i(k) = C_{\xi_i} \mathbf{x}_{\xi_i}(k), \quad (5)$$

with A_{ξ_i} satisfying a polynomial difference equation (polynomial/periodic signals), and $\xi_{\text{res}}(k)$ denoting a residual, unstructured disturbance component.

Problem 1 (State and disturbance estimation). Given Assumption 1, design an observer that provides estimates $\hat{\mathbf{x}}_p(k)$ and $\hat{\xi}(k)$ such that, in the nominal case (without $\xi_{\text{res}}(k)$),

$$\lim_{k \rightarrow \infty} \|\mathbf{x}_p(k) - \hat{\mathbf{x}}_p(k)\| = 0, \quad \lim_{k \rightarrow \infty} |\xi(k) - \hat{\xi}(k)| = 0.$$

In the presence of $\xi_{\text{res}}(k)$, the estimation errors remain bounded and are attenuated.

Let $y^*(k)$ denote a smooth reference signal and define the tracking error $e(k) = y(k) - y^*(k)$.

Problem 2 (Tracking with disturbance rejection). Under Assumption 1 and based on the estimates provided by Problem 1, design a control law: $u(k) = \kappa(\hat{\mathbf{x}}_p(k), \hat{\xi}(k), y^*(k))$ such that, in the nominal case, $\lim_{k \rightarrow \infty} e(k) = 0$, and, in the presence of $\xi_{\text{res}}(k)$, the tracking error $e(k)$ remains bounded and is suitably attenuated.

In the following section, the classical design of an extended state observer with multiple internal models is briefly reviewed. This will, on the one hand, allow us to identify the challenges associated with the classical use of the ESO and, on the other hand, provide an initial comparison with the proposed decoupled observer scheme.

3. Classical extended state observer with multiple internal models

In pursuit of high performance in control systems, it is essential to devise control strategies capable of tracking and rejecting various types of disturbances. One approach to achieve this is through a control structure based on an extended state observer that incorporates multiple internal models. This classic design method will serve as a benchmark for comparisons with the decoupled proposal of this work, particularly

from the perspectives of high dimensionality, complexity, and structural flexibility.

Consider the linear base model given by Eq. (3), perturbed by the disturbance $\xi(k)$ as delineated in Eq. (4). This disturbance is modeled as a composition of various additive components, each potentially stemming from a specific internal model described by a set of difference equations.

3.1. Model structure

For each structured component of the disturbance, $\xi_i(k)$, we assume it adheres to the following state-space representation, characterizing its internal model:

$$\mathbf{x}_{\xi_i}(k+1) = A_{\xi_i} \mathbf{x}_{\xi_i}(k), \quad (6)$$

$$\xi_i(k) = C_{\xi_i} \mathbf{x}_{\xi_i}(k), \quad (7)$$

where $i \in \{1, \dots, r\}$ indexes the set of considered disturbance components. These internal models are integral to the design of the disturbance observer.

The classical approach to designing an extended state observer (ESO) involves augmenting the system model to include these internal models of the disturbance. The augmented system can then be expressed as:

$$\mathbf{x}_p(k+1) = A_p \mathbf{x}_p(k) + B_p \left(u(k) + \sum_{i=1}^r C_{\xi_i} \mathbf{x}_{\xi_i}(k) + \xi_{\text{res}}(k) \right), \quad (8)$$

$$\mathbf{x}_{\xi_i}(k+1) = A_{\xi_i} \mathbf{x}_{\xi_i}(k) \quad \text{for each } i \in \{1, \dots, r\}, \quad (9)$$

$$y(k) = C_p \mathbf{x}_p(k). \quad (10)$$

It is worth noting that the residual term $\xi_{\text{res}}(k)$ in Eq.(8) is not explicitly modelled in the observer. This component represents an unstructured part of the disturbance for which no internal model is available; therefore, it cannot be reconstructed through additional observer states. In the observer design, $\xi_{\text{res}}(k)$ acts as an unknown input entering through the input channel and is treated as an exogenous disturbance to be attenuated, while only the structured components $\xi_i(k)$, $i = 1, \dots, r$, are explicitly estimated via their associated internal models. Nevertheless, if the residual disturbance $\xi_{\text{res}}(k)$ is a bounded and sufficiently smooth signal, the presence of this signal does not represent a risk for the control strategy as all the set of observers will attenuate but not eliminate the residual disturbance $\xi_{\text{res}}(k)$.

3.2. Observer design

An observer for the augmented system based on the output $y(k)$ is structured as follows:

$$\hat{\mathbf{x}}_p(k+1) = A_p \hat{\mathbf{x}}_p(k) + B_p \left(u(k) + \sum_{i=1}^r C_{\xi_i} \hat{\mathbf{x}}_{\xi_i}(k) \right) + L_p(y(k) - \hat{y}(k)), \quad (11)$$

$$\hat{\mathbf{x}}_{\xi_i}(k+1) = A_{\xi_i} \hat{\mathbf{x}}_{\xi_i}(k) + L_{\xi_i}(y(k) - \hat{y}(k)), \quad i = 1, \dots, r, \quad (12)$$

$$\hat{y}(k) = C_p \hat{\mathbf{x}}_p(k). \quad (13)$$

Defining the comprehensive observer states as:

$$\mathbf{x}(k) = \begin{bmatrix} \mathbf{x}_p(k) \\ \mathbf{x}_{\xi_1}(k) \\ \vdots \\ \mathbf{x}_{\xi_r}(k) \end{bmatrix}, \quad \hat{\mathbf{x}}(k) = \begin{bmatrix} \hat{\mathbf{x}}_p(k) \\ \hat{\mathbf{x}}_{\xi_1}(k) \\ \vdots \\ \hat{\mathbf{x}}_{\xi_r}(k) \end{bmatrix}, \quad (14)$$

the observer takes the following form:

$$\hat{\mathbf{x}}(k+1) = (A - LC)\hat{\mathbf{x}}(k) + Bu(k) + Ly(k), \quad (15)$$

where A , B , and C are appropriately defined block matrices to encapsulate the dynamics of the plant and disturbance models:

$$A = \begin{bmatrix} A_p & B_p C_{\xi_1} & \cdots & B_p C_{\xi_r} \\ 0 & A_{\xi_1} & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & & A_{\xi_r} \end{bmatrix}, \quad B = \begin{bmatrix} B_p \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

$$C = [C_p \quad 0 \quad \cdots \quad 0], \quad L = \begin{bmatrix} L_p \\ L_{\xi_1} \\ \vdots \\ L_{\xi_r} \end{bmatrix}.$$

where L is the observer parameter vector.

3.3. Challenges and observations

The classical ESO with multiple internal models provides a unified framework to estimate plant states and structured disturbance components. However, the aggregation of all internal models into a single augmented state leads to high observer order, potential numerical ill-conditioning in the gain computation, and a lack of structural flexibility. These aspects motivate the decoupled multi-observer structure proposed in the next section.

3.3.1. Large-dimension problem

Consider the ESO observer in Eq.(15), where the eigenvalues of $\tilde{A} = A - LC$ are placed inside the unit circle to ensure asymptotic convergence of the estimation error. When the number of disturbance components to be rejected increases, the augmented matrix A grows in dimension, and the gain vector L must be computed for a high-order system.

A standard design technique is Ackermann's formula (Ackermann, 1972),

$$L^T = [0 \quad \cdots \quad 0 \quad 1](\mathcal{O}^T)^{-1} \Delta_{\tilde{A}}(A^T), \quad (16)$$

where $\Delta_{\tilde{A}}(\cdot)$ is the desired characteristic polynomial and $\mathcal{O}$ is the Kalman observability matrix (Kalman et al., 1960). Since this computation requires the inversion of $\mathcal{O}$, a well-conditioned observability matrix is essential.

To illustrate the impact of including multiple internal models, consider an arbitrary second-order plant:

$$x(k+1) = \begin{bmatrix} -1.61 & -0.648 \\ 1 & 0 \end{bmatrix} x(k) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(k),$$

$$y(k) = [1 \quad 0.5] x(k),$$

subject to a disturbance of the form

$$\xi(k) = \alpha_0 H(k) + \sum_{i=1}^m \alpha_i \sin(p_i \omega k + \phi_i),$$

where $H(k)$ is the Heaviside step function and p_i denotes the i th prime number. The corresponding internal models lead to additional states with matrices

$$A_{\xi_1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \quad C_{\xi_1} = [1 \quad 0],$$

$$A_{\xi_{i+1}} = \begin{bmatrix} 0 & 1 \\ -1 & \cos(p_i \omega T) \end{bmatrix}, \quad C_{\xi_{i+1}} = [1 \quad 0], \quad i = 1, \dots, m.$$

The conditioning of the pole-placement problem is evaluated via the condition number (Belsley et al., 1981), $\kappa(\mathcal{O}) = \|\mathcal{O}^{-1}\| \|\mathcal{O}\|$, (using the 2-norm), where large values indicate an ill-conditioned linear system for $Ax = b$. As shown in Fig. 1, the condition number of the observability matrix grows rapidly (on a logarithmic scale) as the number of internal models m increases, indicating that the ESO design becomes progressively more ill-conditioned for a larger number of disturbance components.

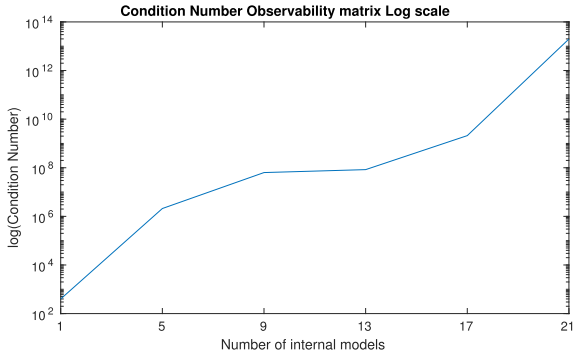


Fig. 1. Condition number of the observability matrix as the number of internal models increases.

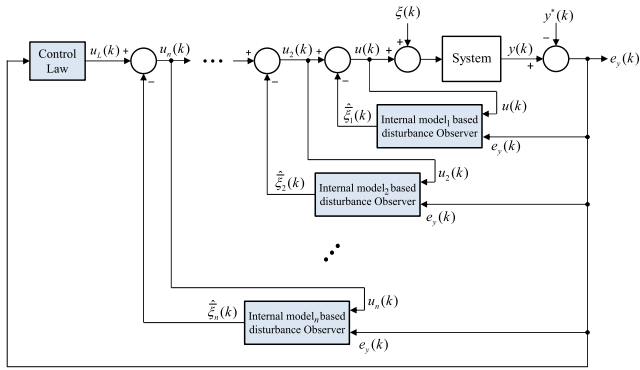


Fig. 2. Decoupled observer structure for n-stages.

3.3.2. High-gain problem

High-gain observers typically place the eigenvalues of $(A - LC)$ with a bandwidth large enough to capture the relevant dynamics. For high-order augmented systems, this often leads to very high gains, with well-known drawbacks such as peaking phenomena and increased sensitivity to noise and modelling errors. When a single ESO is tasked with estimating all disturbance components, it may employ unnecessarily high bandwidth for some signals that would only require moderate gains, resulting in a conservative and numerically fragile design.

3.3.3. Structural rigidity

The single ESO structure also lacks structural flexibility. Once an observer has been tuned to estimate and reject r disturbance components, the inclusion of an additional component $(r + 1)$ generally requires a complete redesign of the augmented observer, with the associated numerical and high-gain issues. This rigidity is particularly limiting in applications where the disturbance description evolves over time.

These challenges motivate the multi-observer framework introduced in the next section, in which each internal model is associated with a dedicated observer, leading to lower-order designs, improved numerical properties, and easier reconfiguration when new disturbance components need to be considered.

4. Decoupled observers scheme

The proposed approach is applicable to both linear and nonlinear differentially flat systems. The decoupled observers scheme employs the nominal design base model of the plant to construct a series of tandem observers. Each observer is dedicated to handling a specific type of internal model, associated with either disturbance rejection or trajectory tracking. This design allows for the incorporation of multiple decoupled observers, each tailored to a specific internal model. Fig. 2 illustrates the schematic for an n -stage decoupled observer system.

The essence of the decoupled scheme lies in its ability to actively reject components of the unified disturbance signal that correspond to the internal model of each specific observer. The disturbance estimate, equivalent to the control signal of the first stage, is subtracted from the control input to produce an auxiliary control signal u_2 . This signal is then utilized by the subsequent observer in the sequence. All following observers receive the tracking error as their input.

4.1. Staged design of decoupled observers

The design of the decoupled observers is phased, with each stage addressing a specific internal model. Importantly, there is no fixed order for addressing the internal models; however, conceptually, it may be advantageous to assign the last stage to handle the polynomial internal model. This model generally accommodates any residual disturbances effectively.

Typically, the first stage is assigned to the internal model or component $\bar{\xi}_1$ of the total disturbance. Subsequent stages sequentially address the remaining components $\bar{\xi}_2, \bar{\xi}_3$, etc., following the order established in the design phase.

The stages incorporate tracking error analysis, denoted as $e_y(k) = y(k) - y^*(k)$, and involves transforming the state vector of the design base model into a state vector of tracking errors. This transformation is achieved by considering the following unperturbed nominal model:

$$\mathbf{x}_p^*(k+1) = A_p \mathbf{x}_p^*(k) + B_p u^*(k), \quad (17)$$

$$y^*(k) = C_p \mathbf{x}_p^*(k). \quad (18)$$

Subtracting this nominal model from the disturbed model given in Eq. (3) yields:

$$\mathbf{x}_p(k+1) - \mathbf{x}_p^*(k+1) = A_p [\mathbf{x}_p(k) - \mathbf{x}_p^*(k)] + B_p [u(k) + \xi(k) - u^*(k)], \quad (19)$$

$$e_y(k) = C_p [\mathbf{x}_p(k) - \mathbf{x}_p^*(k)]. \quad (20)$$

For simplicity in subsequent designs, we define:

$$\bar{\mathbf{x}}_p(k) = \mathbf{x}_p(k) - \mathbf{x}_p^*(k), \quad (21)$$

$$\bar{\xi}(k) = \xi(k) - u^*(k). \quad (22)$$

With these definitions, the dynamics of the error state can be modeled as:

$$\bar{\mathbf{x}}_p(k+1) = A_p \bar{\mathbf{x}}_p(k) + B_p [u(k) + \bar{\xi}(k)], \quad (23)$$

$$e_y(k) = C_p \bar{\mathbf{x}}_p(k). \quad (24)$$

Here, $\bar{\xi}(k)$ encompasses both the effects previously included in the disturbance ξ and the control input $u^*(k)$ of the reference signal. Consequently, the reference signal y^* effectively becomes part of the unified disturbance signal $\bar{\xi}$.

This disturbance is conceptualized as the sum of several additive components, each associated with a specific internal model, regardless of the origins of its components—be they external perturbations, unmodeled dynamics, or reference terms:

$$\bar{\xi}(k) = \bar{\xi}_1(k) + \bar{\xi}_2(k) + \dots + \bar{\xi}_r(k) + \bar{\xi}_{\text{res}}, \quad (25)$$

where $\bar{\xi}_{\text{res}}$ represents the residual disturbance component. The proposal simplifies the design's structural complexity by separating each required observer and feeding back the disturbance estimation state only, highlighting the novelty of the work presented.

4.2. First stage design

The first observer stage in our scheme is based on an augmented model that incorporates the state $\bar{\mathbf{x}}_p$ and a state for the disturbance $\bar{\xi}_1$, associated with the first internal model. The difference equation for this model's component $\bar{\xi}_1$ applied to an arbitrary disturbance $\bar{\xi}_x$ is:

$$p_{\bar{\xi}_1}(\bar{\xi}_x, k) = \bar{\xi}_x(k + m_1) + \gamma_{m_1-1} \bar{\xi}_x(k + m_1 - 1) + \dots + \gamma_0 \bar{\xi}_x(k), \quad (26)$$

where m_1 is the degree of the difference equation of disturbance component $\bar{\xi}_1$, and it holds that:

$$p_{\bar{\xi}_1}(\bar{\xi}_1, k) = 0. \quad (27)$$

The state variables associated with the disturbance's internal model are structured as follows:

$$\bar{\mathbf{x}}_{\bar{\xi}_1}(k) = \begin{bmatrix} \bar{\xi}(k) \\ \bar{\xi}(k+1) \\ \vdots \\ \bar{\xi}(k+m_1-1) \end{bmatrix}, \quad (28)$$

leading to the state-space model:

$$\bar{\mathbf{x}}_{\bar{\xi}_1}(k+1) = A_{\bar{\xi}_1} \bar{\mathbf{x}}_{\bar{\xi}_1}(k) + F_{\bar{\xi}_1} p_{\bar{\xi}_1}(\bar{\xi}, k), \quad (29)$$

$$\bar{\xi}(k) = C_{\bar{\xi}_1} \bar{\mathbf{x}}_{\bar{\xi}_1}(k). \quad (30)$$

In this structure:

$$A_{\bar{\xi}_1} = \begin{bmatrix} 0 & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \\ 0 & \dots & 0 & 1 \\ -\gamma_0 & -\gamma_1 & \dots & -\gamma_{m_1-1} \end{bmatrix}, \quad F_{\bar{\xi}_1} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, \quad \text{and} \quad C_{\bar{\xi}_1} = \begin{bmatrix} 1 & 0 & \dots & 0 \end{bmatrix}.$$

In this augmented model, the term $F_{\bar{\xi}_1} p_{\bar{\xi}_1}(\bar{\xi}, k)$ refers to the application of Internal Model 1 to the total disturbance $\bar{\xi}$, which generates a residual disturbance. This residual disturbance acts as the disturbance for Stage 2 and will be instrumental in determining the estimation error of the disturbance in Stage 1.

Defining $\bar{\mathbf{x}}_1(k) = \begin{bmatrix} \bar{\mathbf{x}}_{\bar{\xi}_1}(k) \\ \bar{\xi}_1(k) \end{bmatrix}$, the augmented model is:

$$\bar{\mathbf{x}}_1(k+1) = A_1 \bar{\mathbf{x}}_1(k) + B_1 u(k) + F_1 p_{\bar{\xi}_1}(\bar{\xi}, k), \quad (31)$$

$$e_y(k) = C_1 \bar{\mathbf{x}}_1(k), \quad (32)$$

$$\text{with } A_1 = \begin{bmatrix} A_p & B_p C_{\bar{\xi}_1} \\ 0 & A_{\bar{\xi}_1} \end{bmatrix}, B_1 = \begin{bmatrix} B_p \\ 0 \end{bmatrix}, C_1 = \begin{bmatrix} C_p & 0 \end{bmatrix} \text{ and } F_1 = \begin{bmatrix} 0 \\ F_{\bar{\xi}_1} \end{bmatrix}.$$

Based on the augmented system model Eq.(31), the following Luenberger-like extended-state observer structure is proposed to provide a estimation of the first component of the disturbance signal $\bar{\xi}$, i.e. $\hat{\bar{\xi}}_1$:

$$\hat{\mathbf{x}}_1(k+1) = A_1 \hat{\mathbf{x}}_1(k) + B_1 u(k) + L_1 (e_y(k) - \hat{e}_{y,1}(k)) \quad (33)$$

$$\hat{e}_{y,1}(k) = C_1 \hat{\mathbf{x}}_1(k) \quad (34)$$

$$\hat{\bar{\xi}}_1(k) = \begin{bmatrix} 0_{1 \times n} & C_{\bar{\xi}_1} \end{bmatrix} \hat{\mathbf{x}}_1(k) \quad (35)$$

$$\text{with } L_1 = \begin{bmatrix} l_0 & \dots & l_{n+m_1-1} \end{bmatrix}^T.$$

The following vector of estimation errors is defined:

$$\bar{\mathbf{x}}_1(k) = \bar{\mathbf{x}}_1(k) - \hat{\mathbf{x}}_1(k) \quad (36)$$

from Eqs.(31) and (33) we obtain the dynamics of the estimation error vector:

$$\bar{\mathbf{x}}_1(k+1) = A_{o,1} \bar{\mathbf{x}}_1(k) + F_1 p_{\bar{\xi}_1}(\bar{\xi}, k), \quad (37)$$

with $A_{o,1} = (A_1 - L_1 C_1)$, where L_1 should be selected such that the eigenvalues of the matrix $A_{o,1}$ are inside the unitary circle to ensure stability.

4.2.1. Estimation error dynamics

Under the presumption of accurate disturbance estimation, stability is secured by the adequate placement of the closed-loop poles, characterizing a classic linear control problem framework.

The dynamics of estimation errors, especially those associated with the disturbance observed in the first stage, are of paramount interest. Through an analysis of these errors, it is confirmed that components complying with the internal model are effectively nullified.

Consider the $\mathcal{Z}$ -transform of the estimation error dynamics Eq.(37), assuming zero initial conditions:

$$z \tilde{\mathbf{X}}_1(z) = A_{o,1} \tilde{\mathbf{X}}_1(z) + F_1 \bar{p}_{\bar{\xi}_1}(z) \tilde{\Xi}(z), \quad (38)$$

where $\bar{p}_{\bar{\xi}_1}(z) = z^{m_1} + \gamma_{m_1-1} z^{m_1-1} + \dots + \gamma_1 z + \gamma_0$ (see (26)), and $\mathcal{Z}\{\bar{\xi}(k)\} = \tilde{\Xi}(z)$.

The estimation error vector's dynamics in the $\mathcal{Z}$ -domain are then expressed by:

$$\tilde{\mathbf{X}}_1(z) = (zI - A_{o,1})^{-1} F_1 \bar{p}_{\bar{\xi}_1}(z) \tilde{\Xi}(z), \quad (39)$$

indicating the comprehensive application of the internal model operator $\bar{p}_{\bar{\xi}_1}(z)$ to the disturbance. Specifically, for disturbance estimation:

$$\begin{aligned} \tilde{\Xi}(z) - \hat{\tilde{\Xi}}_1(z) &= \begin{bmatrix} 0_{1 \times n} & C_{\bar{\xi}_1} \end{bmatrix} \tilde{\mathbf{X}}_1(z) \\ &= \begin{bmatrix} 0_{1 \times n} & C_{\bar{\xi}_1} \end{bmatrix} (zI - A_{o,1})^{-1} F_1 \bar{p}_{\bar{\xi}_1}(z) \tilde{\Xi}(z), \end{aligned} \quad (40)$$

suggesting that the estimation error dynamics are fundamentally influenced by the internal model #1.

To simplify this expression and effectively analyze the dynamics of the estimation error, consider that:

$$\begin{bmatrix} 0_{1 \times n} & C_{\bar{\xi}_1} \end{bmatrix} (zI - A_{o,1})^{-1} F_1 = \frac{N_1(z)}{\det(zI - A_{o,1})}, \quad (41)$$

where $N_1(z)$ is the resulting polynomial from the proposed operation, therefore:

$$\tilde{\Xi}_1(z) := \tilde{\Xi}(z) - \hat{\tilde{\Xi}}_1(z) = \frac{N_1(z)}{\det(zI - A_{o,1})} \bar{p}_{\bar{\xi}_1}(z) \tilde{\Xi}(z). \quad (42)$$

Noteworthy observations about the estimation error dynamics include:

- The denominator of the related transfer function is inherently stable, as ensured by the design of the observing system.
- The difference operator $\bar{p}_{\bar{\xi}_1}(z)$ effectively removes any components of $\tilde{\Xi}(z)$ that conform to the internal model of $\bar{\xi}_1$.

Remark 3. In the specific case of polynomial disturbances of degree $r-1$, these disturbances correspond to an internal model given by $\bar{p}_r(z) = (z-1)^r$, which can be directly used for designing the corresponding observer as presented in this section. However, due to the intrinsic decoupling capability of the proposed approach, this design can be segmented into r identical stages, each structured using the first-degree internal model polynomial: $\bar{p}_1(z) = (z-1)^1$, which corresponds to the annihilator of constant signals.

This segmentation is justified by the estimation error derived in each stage, Eq.(40), which, in this case, is related to the z -domain operator $(z-1)$, performing a difference operation on the current disturbance. The estimation error of the equivalent disturbance in stage i , resulting from this difference operator on the current equivalent disturbance, serves as the new equivalent disturbance signal for the subsequent stage (stage $i+1$, with $i=1, \dots, r$). The nested and decoupled handling of this difference operation ultimately produces the desired effect of $(z-1)^r$, effectively rejecting the polynomial disturbance.

4.3. Second stage design

The observer design for the second stage adheres to the foundational principles established in the first stage, utilizing the same nominal plant model. In this stage, the control input is defined by the relationship:

$$u_1 = u_2 - \hat{\bar{\xi}}_1, \quad (43)$$

where, in general for any stage i , $u_i = u_{i+1} - \hat{\bar{\xi}}_i$ as illustrated in Fig. 2.

The input signals for the second observer are the tracking error e_y and the adjusted control input u_2 . Importantly, the design does not prescribe a specific sequential order for processing the internal models. Consequently, the internal models and their respective observers can be arbitrarily numbered and configured without adherence to a predetermined sequence.

4.4. Dynamics of the disturbance estimation error

According to the structure proposed for the scheme of decoupled observers, each stage provides an estimation subtracted from the current disturbance signal (see Fig. 2). For the first stage, this difference corresponds to the disturbance estimation error calculated in Eq.(42). This first error acts as the current disturbance (residual disturbance) that must be considered for the second stage. Following a development similar to that shown in Section 4.2.1, it is straightforward to obtain the current disturbance estimation error for the second stage as:

$$\tilde{\Xi}_2(z) = \tilde{\Xi}_1(z) - \hat{\Xi}_2(z) = \frac{N_2(z)}{\det(zI - A_{o,2})} \bar{p}_{\xi_2}(z) \tilde{\Xi}_1(z). \quad (44)$$

Similarly, for the third stage, it results in:

$$\tilde{\Xi}_3(z) = \tilde{\Xi}_2(z) - \hat{\Xi}_3(z) = \frac{N_3(z)}{\det(zI - A_{o,3})} \bar{p}_{\xi_3}(z) \tilde{\Xi}_2(z). \quad (45)$$

If r stages are considered, for the r -th stage, we have:

$$\tilde{\Xi}_r(z) = \tilde{\Xi}_{r-1}(z) - \hat{\Xi}_r(z) = \frac{N_r(z)}{\det(zI - A_{o,r})} \bar{p}_{\xi_r}(z) \tilde{\Xi}_{r-1}(z). \quad (46)$$

Replacing each of the preceding results in the last equation, we find that:

$$\tilde{\Xi}_r(z) = \frac{N_r(z) \dots N_1(z)}{\det(zI - A_{o,r}) \dots \det(zI - A_{o,1})} \left(\bar{p}_{\xi_r}(z) \dots \bar{p}_{\xi_1}(z) \right) \tilde{\Xi}(z). \quad (47)$$

Notable observations concerning this residual disturbance include the following:

- The transfer function $\frac{\tilde{\Xi}_r(z)}{\tilde{\Xi}(z)}$ is stable since its corresponding denominator results from the multiplication of r stable polynomials, each adjusted to the observation system of each stage, which is stable by construction.
- The numerator term $\left(\bar{p}_{\xi_r}(z) \dots \bar{p}_{\xi_1}(z) \right)$ corresponds to the annihilator polynomial of the decoupled scheme that is ultimately applied to the disturbance $\tilde{\Xi}(z)$. Thus, each of the annihilator polynomials corresponding to each considered internal model is applied in a composite manner.
- If the disturbance signal corresponding to $\tilde{\Xi}(z)$ is bounded, the output corresponding to this system, i.e., the disturbance estimation error in the last stage, will also be bounded.

4.5. Control law

The control law can be formulated in two distinct approaches:

1. **Observer-Based Control Law:** Utilizing the estimation of $\mathbf{x}_p$ provided by the final observation stage (r -stage), a simple observer-based linear control law can be implemented (see Fig. 2):

$$u_L(k) = k \hat{\mathbf{x}}_p \quad (48)$$

Here, if the pair (A_p, B_p) is controllable, the eigenvalues of $(A_p - B_p k)$ can be arbitrarily assigned by selecting an appropriate feedback gain k . This ensures that the characteristic polynomial of the tracking error e_y is configured to have all its coefficients within the unit circle, guaranteeing system stability.

2. **Stabilizing Linear Control:** The second approach involves a separate linear control responsible for stabilizing the unperturbed linear system. This method does not rely on the estimated state space vector of the plant. Instead, a control law represented by a transfer function:

$$C(z) = \frac{Num_c(z)}{Den_c(z)} \quad (49)$$

is used, where:

$$U_L(z) = -C(z)E_y(z) \quad (50)$$

The overall control system dynamics are described by:

$$G_o(z) = \frac{C(z)G_p(z)}{1 + C(z)G_p(z)} \quad (51)$$

Here, Num_c and Den_c are chosen algebraically to ensure that the denominator of $G_o(z)$ corresponds to the desired characteristic polynomial of the tracking error.

4.6. Stability analysis

A primary concern regarding the proposed design relates to the stability of the closed-loop system as the number of stages increases. This section intends to illustrate the structural flexibility feature of the proposed scheme based on the stability analysis while including or removing an existing observer from the structure. Stability is guaranteed for each pair of Control law-observer i , assuming accurate disturbance estimation, based on the separation principle (applicable under the assumption of a linearly perturbed system (Levine, 2018)). The addition of an extra stage does not compromise the system's stability. Consider the generalization of Eq. (43), which delineates the relationship between the control input at any stage and the corresponding disturbance estimation.

Then, considering initially one single observer, the stability analysis can be performed as follows.

Theorem 1. Let the observation error system for a stable internal-model-based ESO observer be,

$$\tilde{x}_1(k+1) = A_{o,1} \tilde{x}_1 + F_1 p_{\xi_1}(\tilde{\xi}, k)$$

Then, if $\xi(k)$ is a Lipschitz function constituted by the endogenous dynamics, the observation error system is asymptotically stable.

Proof. For each non-periodic disturbance term, an ESO observer is proposed with the error dynamics defined in Eq.(37) and given by:

$$\tilde{x}_1(k+1) = A_{o,1} \tilde{x}_1 + F_1 p_{\xi_1}(\tilde{\xi}, k)$$

With $A_{o,1} = (A_1 - L_1 C_1)$. The individual observer's stability is secured by the proper selection of L_1 such that the eigenvalues of the matrix $A_{o,1}$ are inside the unit circle of the complex plane. If this is satisfied then exists an unique $P_1 > 0$ such that, given $Q_1 > 0$, Lyapunov's equation $A_{o,1}^T P_1 A_{o,1} - P_1 + Q_1 = 0$, is satisfied. Then, the following Lyapunov function candidate $V(\tilde{x}_1) = \tilde{x}_1^T P_1 \tilde{x}_1$ definite positive is used and $\Delta V = V(\tilde{x}_1(k+1)) - V(\tilde{x}_1(k))$ is given by:

$$\begin{aligned} \Delta V &= \tilde{x}_1^T \left(A_{o,1}^T P_1 A_{o,1} - P_1 \right) \tilde{x}_1 \\ &\quad + \left(F_1 p_{\xi_1}(\tilde{\xi}, k) \right)^T P_1 \left[F_1 p_{\xi_1}(\tilde{\xi}, k) \right] \\ &= -\tilde{x}_1^T Q_1 \tilde{x}_1 + \left(\left[F_1 p_{\xi_1}(\tilde{\xi}, k) \right]^T P_1 \left[F_1 p_{\xi_1}(\tilde{\xi}, k) \right] \right) \\ &\leq -\lambda_{\max Q_1} \|\tilde{x}_1\| + \left(\left[F_1 p_{\xi_1}(\tilde{\xi}, k) \right]^T P_1 \left[F_1 p_{\xi_1}(\tilde{\xi}, k) \right] \right) \end{aligned}$$

If the disturbance is globally Lipschitz, and remembering that $F_1 p_{\xi_1}(\tilde{\xi}, k)$ depends on $\tilde{x}_1$ for disturbances that lump the nonlinear terms of the system, then it holds that:

$$\left| \left[F_1 p_{\xi_1}(\tilde{\xi}, k) \right]^T P_1 \left[F_1 p_{\xi_1}(\tilde{\xi}, k) \right] \right| \leq \left| \left[F_1 p_{\xi_1}(\tilde{\xi}, k) \right] \right|^2 \leq c_1 \|\tilde{x}_1\|$$

For external disturbances, the only assumptions that can be made are boundness for the disturbance and its derivative, i.e., $\left| F_1 p_{\xi_1}(\tilde{\xi}, k) \right| \leq k_{\tilde{\xi}}$ and $\left| d/dt \left\{ F_1 p_{\xi_1}(\tilde{\xi}, k) \right\} \right| \leq k_{d\tilde{\xi}}$. Therefore, using $\lambda_{\max Q_1}$ as the maximum eigenvalue of Q_1 ,

$$\Delta V \leq -\lambda_{\max Q_1} \|\tilde{x}_1\| + c_1 \|\tilde{x}_1\| \leq -(\lambda_{\max Q_1} - c_1) \|\tilde{x}_1\|$$

Then, the observation error will be asymptotically stable if $\lambda_{\max Q_1} > c_1$. $\square$

Now, if multiple parallel internal-model-based ESO observers are implemented, the stability analysis can be performed as follows.

Theorem 2. *If the disturbance $\xi(k)$ is a state-dependent Lipschitz function and an arbitrary number of stable parallel internal-model-based ESO observers are implemented, then the observation error will be asymptotically stable.*

Proof. Consider an arbitrary number of stable parallel observers v . Then, the following general Lyapunov function candidate can be proposed:

$$V(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_v) = \tilde{x}_1^T P_1 \tilde{x}_1 + \tilde{x}_2^T P_2 \tilde{x}_2 + \dots + \tilde{x}_v^T P_v \tilde{x}_v$$

where each observation error system is defined by:

$$\tilde{x}_i(k+1) = A_{o,i} \tilde{x}_i + F_i p_{\xi_i}(\tilde{\xi}, k)$$

where the term $F_i p_{\xi_i}(\tilde{\xi}, k)$ represents the residual disturbance left by the previous observer $i-1$ ($F_0 p_{\xi_0}(\tilde{\xi}, k) = \xi(k)$) and each observer is stable, i.e., each eigenvalue of $A_{o,i} = (A_i - L_i C_i)$ lies inside the unit complex circle.

Then, $\Delta V = V(\tilde{x}_1(k+1), \tilde{x}_2(k+1), \dots, \tilde{x}_v(k+1)) - V(\tilde{x}_1(k), \tilde{x}_2(k), \dots, \tilde{x}_v(k))$ is,

$$\begin{aligned} \Delta V &= \tilde{x}_1^T \left(A_{o,1}^T P_1 A_{o,1} - P_1 \right) \tilde{x}_1 + \tilde{x}_2^T \left(A_{o,2}^T P_2 A_{o,2} - P_2 \right) \tilde{x}_2 \\ &\quad + \dots + \tilde{x}_v^T \left(A_{o,v}^T P_v A_{o,v} - P_v \right) \tilde{x}_v \\ &\quad + \left(\left[F_1 p_{\xi_1}(\tilde{\xi}, k) \right]^T P_1 \left[F_1 p_{\xi_1}(\tilde{\xi}, k) \right] \right) \\ &\quad + \dots + \left(\left[F_v p_{\xi_v}(\tilde{\xi}, k) \right]^T P_v \left[F_v p_{\xi_v}(\tilde{\xi}, k) \right] \right) \\ &= -\tilde{x}_1^T Q_1 \tilde{x}_1 - \tilde{x}_2^T Q_2 \tilde{x}_2 - \dots - \tilde{x}_v^T Q_v \tilde{x}_v \\ &\quad + \left(\left[F_1 p_{\xi_1}(\tilde{\xi}, k) \right]^T P_1 \left[F_1 p_{\xi_1}(\tilde{\xi}, k) \right] \right) \\ &\quad + \dots + \left(\left[F_v p_{\xi_v}(\tilde{\xi}, k) \right]^T P_v \left[F_v p_{\xi_v}(\tilde{\xi}, k) \right] \right) \end{aligned}$$

Once again, if each component of the disturbance is Lipschitz concerning the state $\tilde{x}_i$ for the disturbances that lump the system's nonlinearities, then each term can be bounded by a constant. For external disturbances, the boundness supposition holds. This follows:

$$\Delta V \leq -(\lambda_{\max Q_1} - c_1) \|\tilde{x}_1\| - \dots - (\lambda_{\max Q_v} - c_v) \|\tilde{x}_v\|$$

□

Suppose that the closed-loop system is already configured with r observer stages and requires an additional observer due to a new disturbance $\xi_{r+1}(k)$, characterized by a known and well-defined internal model.

The disturbance is then defined as:

$$\xi(k) = \xi_1(k) + \xi_2(k) + \dots + \xi_r(k) + \xi_{r+1}(k)$$

Notice that the stability is independent of the number of observers considered, and if an observer is activated or deactivated in the scheme, the stability of the general closed-loop system remains the same, as the contribution of the particular observer is activated or deactivated in the Lyapunov function. This implies a robustness property on the arbitrary inclusion or exclusion of any observer in the scheme.

4.7. Computational cost

A natural concern regarding the proposed scheme is the computational cost of using multiple observers. In the classical implementation, a single ESO aggregates all internal models in an augmented state, leading to a large observer matrix whose dimension grows with the number of disturbance components (see Section 3). In contrast, the present proposal employs several ESOs in parallel, each associated with a single internal model. Hence, each observer remains low-order and only contains

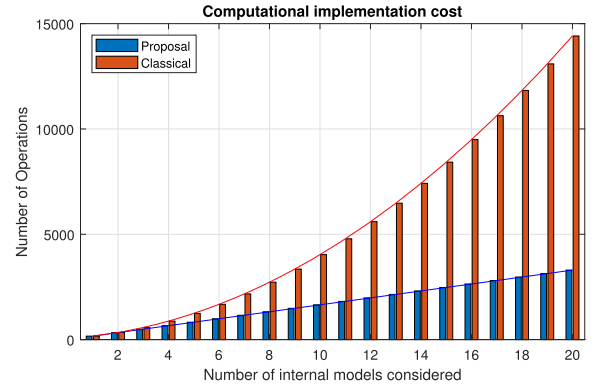


Fig. 3. Computational cost comparison.

the dynamics of one disturbance component. Since there is no directly comparable benchmark for the periodic-signal observer, the following analysis focuses on the ESO-type observers.

To compare both approaches, we estimated the number of arithmetic operations (additions, subtractions and multiplications) required for one sampling step of the observer using the MATLAB System-On-Chip (SoC) analysis function `socfunctionanalyzer`. For each configuration, a function implementing the ESO in the form of Eq.(15) was analysed with random matrices A , B , C and L , considering a plant order $n = 2$ and disturbance internal models of order $n_\xi = 2$. For the classical ESO, the observer dimension is

$$n_{\text{class}} = n + rn_\xi,$$

where r is the number of internal models. For the proposed scheme, each observer has dimension

$$n_{\text{prop}} = n + n_\xi,$$

and r such observers are used in parallel, so the total cost scales with r times the cost of a single observer of dimension n_{prop} .

Fig. 3 shows the estimated number of operations for both schemes, together with trend lines. The proposed multi-observer architecture exhibits an approximately linear growth with the number of internal models, since the dimension of each observer is fixed. In contrast, the classical ESO displays a markedly nonlinear (super-linear) increase in computational cost as r grows, due to the continuously increasing dimension of the augmented state. This comparison reveals an additional benefit of the proposed scheme: a significant reduction in implementation effort, on top of avoiding redundant plant-state estimations.

5. Extension to observer of periodic signals

A straightforward structure for observing periodic signals can be developed by leveraging the duality between control and observation mechanisms. Effective estimation or rejection of a periodic signal is feasible when its corresponding internal model is integrated into the observer design.

Conceptual framework. The fundamental idea is to adapt a traditional control scheme for estimation purposes. In the dual setting used here, the reference signal is the tracking error $e_y(k)$ and the output of the observation system, $\hat{e}_y(k)$, is an estimate of $e_y(k)$.

In the underlying control configuration, when perfect tracking or rejection is achieved, the internal model embedded in the controller generates a control signal $u(k)$ that cancels the periodic disturbance (or reproduces the periodic reference). As a consequence, in steady state $u(k)$ carries, up to a sign, the same periodic content as the original disturbance.

In the dual observer interpretation adopted in this section, this measured signal $u(k)$ is therefore regarded as an *equivalent disturbance input*

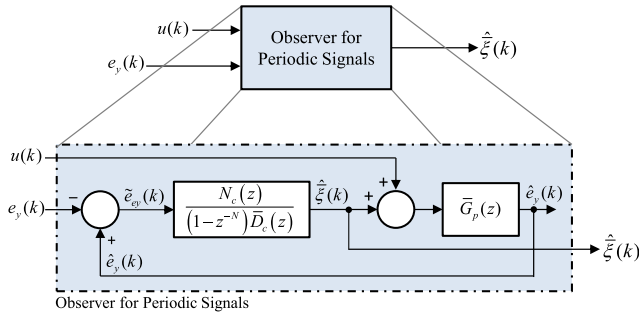


Fig. 4. Schematic representation of the periodic signal observer.

to be estimated. This is the precise sense in which we later state that $u(k)$ “acts as a disturbance” within the observation scheme: physically, $u(k)$ remains the plant input, but for the observer it is the signal that conveys the periodic component to be reconstructed. The resulting observer structure is depicted in Fig. 4.

This approach allows for a seamless integration of periodic signal models into the observation structure, enhancing the system’s ability to track and mitigate the effects of periodic disturbances. Throughout this section we consider, for clarity of exposition, the case in which the system is influenced solely by an equivalent periodic disturbance.

5.1. Analysis as a control scheme

The proposed observer scheme can be analysed by analogy with a control framework. Incorporating the internal model of a periodic signal into a controller equips the closed loop with dual capabilities: tracking periodic references and rejecting periodic disturbances. The observer structure introduced above exploits this duality by reinterpreting the resulting control signal as the equivalent disturbance to be estimated. The detailed analytical development of this structure is given in [Subsection 5.4](#).

5.2. Feedback linearization and linear-like analysis

With a well-designed control system, feedback linearisation (or an equivalent linearising transformation) enables a linear-like analysis of the closed-loop behaviour. The impact of a periodic signal, whether appearing as a reference or as an equivalent disturbance in the control input, manifests in steady state as a periodic pattern (see [Appendix A](#) for further details). Thus, regardless of whether the periodic signal enters through e_y or through u , the controller performs the required tracking or rejection tasks in a unified manner.

5.3. Transient and steady-State behavior

During the transient phase, the error $e_y(k)$ exhibits periodic behaviour, enabling the controller to learn and compensate the periodic component. As the transient vanishes and effective rejection is achieved, $e_y(k) \rightarrow 0$ and the internal model generates a control action that cancels the disturbance. Consequently, the control input $u(k)$ adopts a periodic nature that mirrors the disturbance signal.

In the dual observer interpretation introduced in the conceptual framework, this measured signal $u(k)$ is used as the equivalent disturbance driving the estimation-error dynamics. The observer is therefore designed so that its output $\hat{e}_y(k)$ tracks $e_y(k)$ and, by exploiting the internal model, the estimation error $e_y(k) - \hat{e}_y(k)$ is driven asymptotically to zero while reconstructing the periodic component contained in $u(k)$.

5.4. Observer structure for periodic signals

The observer design for periodic signals employs an input-output approach based on the nominal plant transfer function

$$\bar{G}_p(z) = C_p(zI - A_p)^{-1}B_p, \quad (52)$$

which we write as

$$\bar{G}_p(z) = \frac{\bar{N}_p(z)}{\bar{D}_p(z)} := \frac{\hat{E}_Y(z)}{U(z) + \hat{\Xi}(z)}, \quad (53)$$

where $U(z)$, $E_Y(z)$ and $\hat{E}_Y(z)$ denote the z -transforms of $u(k)$, $e_y(k)$ and $\hat{e}_y(k)$, respectively, and $\hat{\Xi}(z)$ is the transform of the estimated equivalent disturbance.

Thus, the associated difference equation is

$$\bar{D}_p(z)\hat{E}_Y(z) = \bar{N}_p(z)U(z) + \bar{N}_p(z)\hat{\Xi}(z). \quad (54)$$

To focus on the estimation-error dynamics, define $\tilde{E}_{E_Y}(z) = \hat{E}_Y(z) - E_Y(z)$ and subtract $\bar{D}_p(z)E_Y(z)$ from both sides, obtaining

$$\bar{D}_p(z)\tilde{E}_{E_Y}(z) = \bar{N}_p(z)\hat{\Xi}(z) + [\bar{N}_p(z)U(z) - \bar{D}_p(z)E_Y(z)]. \quad (55)$$

Considering the internal model of an N -periodic signal, we apply the corresponding annihilator $(z^N - 1)$:

$$(1 - z^{-N})\bar{D}_p(z)\tilde{E}_{E_Y}(z) = (1 - z^{-N})\bar{N}_p(z)\hat{\Xi}(z) + \dots \\ (1 - z^{-N})\bar{N}_p(z)U(z) - (1 - z^{-N})\bar{D}_p(z)E_Y(z). \quad (56)$$

If we assume that the system is minimum phase (i.e., $\bar{N}_p(z)$ has all its zeros inside the unit circle), we propose the following control law (disturbance estimator):

$$\hat{\Xi}(z) = \frac{1}{(1 - z^{-N})\bar{N}_p(z)} \left[-z^{-N}\bar{D}_p(z)\tilde{E}_{E_Y}(z) - \frac{N_c(z)}{\bar{D}_c(z)}\tilde{E}_{E_Y}(z) \right]. \quad (57)$$

Incorporating this estimator into the dynamics of $\tilde{E}_{E_Y}(z)$ results in

$$[\bar{D}_c(z)\bar{D}_p(z) + N_c(z)]\tilde{E}_{E_Y}(z) = \bar{D}_c(z)(1 - z^{-N})[\bar{N}_p(z)U(z) - \bar{D}_p(z)E_Y(z)]. \quad (58)$$

This expression shows that the dynamics of the estimation error $\tilde{E}_{E_Y}(z)$ are driven by an equivalent disturbance term, which is dynamically attenuated. Suitable polynomials $\bar{D}_c(z)$ and $N_c(z)$ must be selected so that the estimation-error dynamics are dominated by a desired polynomial whose roots lie inside the unit circle of the complex plane.

Remark 4 (Practical implementation with FIR filtering). The internal model of the periodic signal, indirectly represented in the disturbance estimator [Eq. \(57\)](#) by the term $\frac{1}{1-z^{-N}}$, inherently provides infinite gain at the fundamental frequency and all its harmonics. This property can lead to excessive gain at high frequencies, potentially compromising stability and robustness. To mitigate this effect, it is advisable to incorporate a low-pass filter, denoted $H_f(z)$, into the internal model. The modified disturbance estimator for practical implementation becomes

$$\hat{\Xi}(z) = \frac{1}{(1 - z^{-N}H_f(z))\bar{N}_p(z)} \left[-z^{-N}H_f(z)\bar{D}_p(z)\tilde{E}_{E_Y}(z) - \frac{N_c(z)}{\bar{D}_c(z)}\tilde{E}_{E_Y}(z) \right], \quad (59)$$

where $H_f(z)$ is typically designed as a zero-phase finite-impulse-response (FIR) filter with $\|H_f(z)\|_\infty = 1$. The filter is chosen to maintain a high bandwidth while attenuating unwanted high-frequency noise and preserving the fundamental-frequency response.

This structure for estimating periodic disturbances complements the disturbance models considered in the previous sections, providing coverage for a wide spectrum of practically relevant disturbance signals.

FIR filter design. A low-pass, zero-phase FIR filter can be easily designed using the Matlab command `fir1`, balancing filter bandwidth and signal smoothness. The user is expected to tune the bandwidth to effectively mitigate high-frequency noise without excessively damping the periodic estimation. Selecting a very low bandwidth may yield poor estimation performance but strong noise attenuation, whereas a very high bandwidth may improve estimation accuracy at the cost of excessive noise amplification and potential stability issues. Hence, a compromise between noise amplification and estimation performance is required.

6. Proposed methodology

To ensure the proper design of the proposed scheme, the following methodology has been structured. These steps, derived from the theoretical foundations outlined in previous sections, guide all observer stages to ensure their stability, tracking, and rejection accuracy:

1. **Disturbance Identification:** First, identify and characterize the primary disturbance components of the system as described in [Subsection 6.1](#). Next, determine the internal model approximation for each disturbance component. Finally, use these approximations as observer stages.
2. **Design each observer stage:** Begin by designing the stage for polynomial disturbances, followed by the stage for inter-harmonic disturbances, and lastly, the stage for periodic disturbances. The order is flexible, allowing users to choose the sequence that best suits their needs.
 - **Observer for polynomial disturbances:** Define the disturbance matrices $A_{\bar{\xi}_1}$, $C_{\bar{\xi}_1}$ and then determine the augmented matrices for the stage 1:

$$A_1 = \begin{bmatrix} A_p & B_p C_{\bar{\xi}_1} \\ 0 & A_{\bar{\xi}_1} \end{bmatrix}, B_1 = \begin{bmatrix} B_p \\ 0 \end{bmatrix}, C_1 = \begin{bmatrix} C_p & 0 \end{bmatrix}.$$

Then, design the observer gain L_1 for stage 1 by placing the eigenvalues of the matrix $(A_1 - L_1 C_1)$ according to the desired estimation dynamics (see [Remark 3](#)).

Remark 5. If high-order polynomial disturbance rejection or tracking is required, it is recommended that the user employ multiple stages, each configured with a constant disturbance model (i.e. $A_{\bar{\xi}_1} = 1$, $C_{\bar{\xi}_1} = 1$). For example, if ramp disturbance rejection is required, design two identical observer stages with the internal disturbance model set to constant.

- **Observer for inter-harmonic disturbances:** Define the disturbance matrices $A_{\bar{\xi}_2}$, $C_{\bar{\xi}_2}$ for the specific inter-harmonic ω_x :

$$A_{\bar{\xi}_2} = \begin{bmatrix} 0 & 1 \\ -1 & 2 \cos(\omega_x T_s) \end{bmatrix}, C_{\bar{\xi}_2} = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

and determine the augmented matrices for the stage 2:

$$A_2 = \begin{bmatrix} A_p & B_p C_{\bar{\xi}_2} \\ 0 & A_{\bar{\xi}_2} \end{bmatrix}, B_2 = \begin{bmatrix} B_p \\ 0 \end{bmatrix}, C_2 = \begin{bmatrix} C_p & 0 \end{bmatrix}.$$

Then, design the observer gain L_2 for stage 2 by placing the eigenvalues of the matrix $(A_2 - L_2 C_2)$ according to a desired estimation dynamics. If more than one inter-harmonic is identified in the disturbance (e.g. ω_{x_1} , ω_{x_2} , ω_{x_3} , ...), it is recommended that the user employ multiple observer stages, each configured to a different inter-harmonic frequency.

- **Observer for periodic disturbances:** Determine the samples-per-period N characteristic of the periodic disturbance signal. Then, follow the design procedure outlined in [Section 5](#) to design the periodic estimator properly. Additionally, consider the filtering approach highlighted in [Remark 4](#) to manage high gain at high frequencies and enhance robustness.
3. **Control Law:** Design the stabilizing control law as outlined in [Section 4.5](#).

4. **Final integration:** Integrate all observer stages and the stabilizing controller with the system plant following the scheme presented in [Fig. 2](#).

6.1. Characterization of the disturbance signal

The main components of the disturbance signal are typically induced by known external disturbances and reference signals. These suggest the first internal models to be considered in the scheme. In the case of nonlinear differentially flat systems, the total disturbance also includes unmodeled internal dynamics and nonlinear behaviors of endogenous origin. However, once a dominant linear behavior is imposed via feedback and an accurate disturbance estimate is available, the most relevant steady-state components of the total disturbance are those inherited from the primary reference and external signals. Under these conditions, the steady-state response of a stable linear system excited by polynomial, sinusoidal or periodic inputs preserves the same signal class (e.g. a sinusoidal input produces a sinusoidal output at the same frequency); see [Appendix A](#) for details.

For unknown dynamics, or disturbance components that are endogenously generated and not initially contemplated, additional identification procedures can be used to infer suitable internal models. Examples include:

- Spectral analysis (e.g. Fourier transform) of measured disturbance signals, which allows extracting dominant harmonics to be embedded in the observation scheme, see [Ramos Fuentes et al. \(2015\)](#).
- Sparse Identification of Nonlinear Dynamics (SINDy) approaches to identify governing nonlinear dynamics from data, as in [Shangjie Frank Ma and Sun \(2022\)](#).
- Machine-learning-based methods that adaptively configure internal models for use in the observer structure, see [Campos Ordoñez and Ferguson \(2023\)](#).

The performance of the proposed strategy is ultimately limited by the available knowledge of the signals involved. For unmodeled components (but still driven by the primary reference and disturbance signals), a polynomial internal model should always be included, at least with an extended state for constant disturbances, to guarantee a basic level of tracking and rejection for signals not explicitly represented in the internal models. In this sense, classical ADRC (with an ESO containing a single extended state) is well known for effectively rejecting low- and mid-frequency disturbances, albeit at the cost of relatively high observer gains. The proposed decoupled multi-observer scheme, in contrast, achieves comparable disturbance rejection while avoiding excessively high gains, which facilitates implementation in practice.

7. Case study

This case study seeks to demonstrate the practical application and effectiveness of the proposed control scheme in a mechatronic system. The system in question consists of two directly coupled DC motors with a magnetic setup that introduces a periodic load torque (see [Fig. 5](#)).

The primary goal is to control the angular velocity ω of the main motor to achieve the desired performance characteristics, despite disturbances introduced by the load torque and the magnetic setup. The secondary motor, controlled by a current control loop, acts as a generator of the load torque disturbance, creating a dynamic environment to test the robustness and performance of the control scheme.

7.1. System model and problem formulation

The dynamic behavior of the mechatronic system under study is governed by the following differential equation:

$$\dot{y}(t) = -\frac{k_i k_b}{J R_a} y(t) - \frac{b}{J} y(t) + \frac{2\pi k_i k_h}{J R_a} u(t) - \frac{2\pi}{J} \tau_d(t, \theta) - \frac{2\pi}{J} \tau_{ext}(t), \quad (60)$$

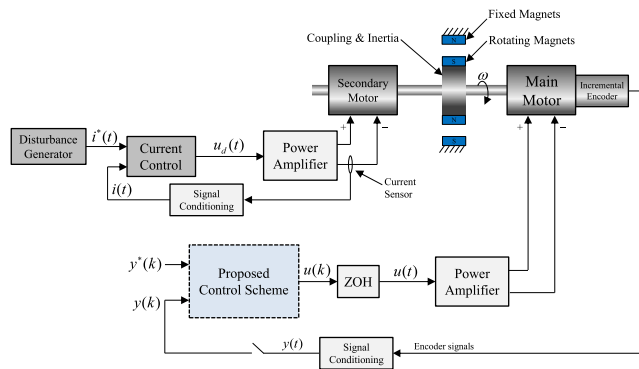


Fig. 5. General setup for the case study, illustrating the mechatronic system with two directly coupled DC motors and a magnetic setup for generating periodic load torque disturbances.

where $y(t)$ is the angular velocity of the motor in rev/s, $u(t)$ is the control input in percentage with range $[-100, +100]$, k_h is the gain of the full-bridge driver $[V/\%]$, k_i is the torque constant, k_b is the back-emf constant, R_a is the armature resistance, J is the equivalent inertia of the system, b is the equivalent viscous-friction of the system, $\tau_d(t, \theta)$ is the non-linear periodic torque produced by the magnetic setup (Cortés-Romero et al., 2014), and $\tau_{ext}(t)$ is the external disturbance torque produced by the secondary motor.

Therefore, following an experimental identification process, the model and its corresponding disturbance input can be expressed as:

$$\dot{y}(t) = - \underbrace{\left(\frac{k_i k_b}{J R_a} + \frac{b}{J} \right)}_{1.164} y(t) + \underbrace{\left(\frac{2\pi k_i k_h}{J R_a} \right)}_{1.0591} (u(t) + \xi(t)) \quad (61)$$

with $\xi(t) = \frac{JR_d}{2\pi k_f k_h} \left(-\frac{2\pi}{J} \tau_d(t, \theta) - \frac{2\pi}{J} \tau_{ext}(t) \right)$. Consequently, discretizing the system via the ZOH method with a sampling time of $T_s = 5$ ms, and deriving its state-space representation, yields:

$$x_p(k+1) = a_p x_p(k) + b_p(u(k) + \xi(k)), \quad (62)$$

$$y(k) = c_p x_p(k), \quad (63)$$

where $a_p = .9942$, $b_p = .00528$, $c_p = 1$, $\xi(k)$ is the discrete-time equivalent input disturbance, incorporating both $\tau_d(t, \theta)$ and $\tau_{ext}(t)$ effects, and $x(k) = y(k)$ representing the angular velocity of the main motor.

7.2. Experimental setup

The experimental platform integrates a mechatronic system and a control prototyping system operating with Matlab in real-time. The system features two directly coupled DC motors, augmented by a magnetic configuration that induces a periodic load torque at constant angular velocities (refer to Fig. 5). This arrangement not only validates the practical viability of the control strategy but also yields critical insights into its real-time performance, effectively narrowing the divide between theoretical innovation and practical deployment.

The primary goal is to control the angular velocity, $y(k)$, of the primary motor to attain the stipulated performance parameters, in spite of the disturbances emanating from the load torque and magnetic assembly. The secondary motor imposes an external load torque on the system, governed via a current control loop. The periodic disturbance emerges from a specific magnet arrangement: two magnets affixed to the rotating axis and another pair positioned on a static frame. As the rotating magnets encounter the stationary ones, alternating forces of repulsion or attraction are experienced, thereby generating a pulsating torque along the axis of the primary motor. Consequently, the aggregate load torque principally comprises a disturbance instigated by the secondary motor's current control loop and a periodic disturbance element introduced by the magnetic configuration.

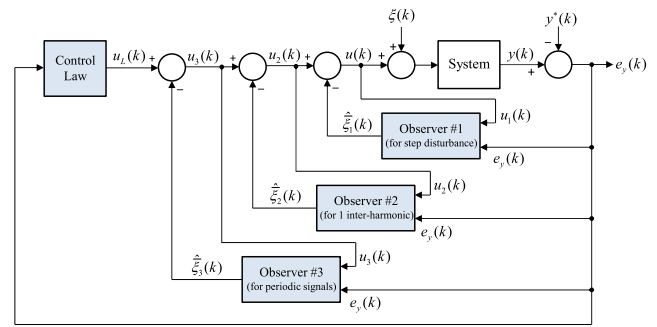


Fig. 6. General scheme for the case study.

7.3. Proposed control scheme

In this experimental case study, the electromechanical setup of Fig. 5 is described by an identified discrete-time linear model that matches the structure Eq.(3). Hence, no explicit feedback-linearisation stage is required in the implementation of the proposed observers and control law.

The proposed control scheme is structured into three distinct stages, each featuring decoupled observers and a stabilizing control law. The initial stage incorporates an Extended-State Observer (ESO) that approximates the internal model of disturbances as a constant value. The subsequent stage employs an ESO that models the disturbance as an interharmonic with a frequency of ω_x rad/s. The final stage implements the innovative observer designed for periodic signals with a fundamental period of T_p secs (see [Section 5](#)). [Fig. 6](#) illustrates the complete architecture of the proposed scheme, including all three observer stages and the stabilizing control law.

Therefore, since the proposed scheme relies entirely on the system represented in the tracking error form, the system in Eqs. (62)-(63) is transformed into the following error-based form:

$$\bar{x}_p(k+1) = a_p \bar{x}_p(k) + b_p(u(k) + \bar{\xi}(k)), \quad (64)$$

$$e_y(k) = c_p \bar{x}_p(k), \quad (65)$$

where $e_v(k) = y(k) - y^*(k)$, and the disturbance $\bar{\xi}(k)$ can be expressed as:

$$\bar{\xi}(k) = \bar{\xi}_1(k) + \bar{\xi}_2(k) + \bar{\xi}_3(k) + \bar{\xi}_{res}(k), \quad (66)$$

where, $\bar{\xi}_1(k)$ contains a constant disturbance, $\bar{\xi}_2(k)$ contains an inter-harmonic disturbance of frequency $\omega_x = 17$ rad/s, $\bar{\xi}_3(k)$ is a T_p -periodic disturbance and $\bar{\xi}_{res}(k)$ represents a residual signal.

7.3.1. Observer 1

Consider the internal model of a constant disturbance $\xi_1(k)$, that emulates a sudden change of load, represented in state-space as:

$$\bar{\mathbf{x}}_{\xi_1}(k+1) = A_{\xi_1} \bar{\mathbf{x}}_{\xi_1}(k) \quad (67)$$

$$\bar{\xi}_1(k) = C_{\bar{\xi}_1} \bar{\mathbf{x}}_{\bar{\xi}_1}(k) \quad (68)$$

with, $A_{\bar{\xi}_1} = 1$, $C_{\bar{\xi}_1} = 1$, $\bar{x}_{\bar{\xi}_1}(k) = \bar{\xi}_1(k)$. Then, for the 1st stage, the augmented system containing the system state $\bar{x}_p(k)$ and the disturbance state $\bar{\xi}_1(k)$ is given by:

$$\bar{\mathbf{x}}_1(k+1) = A_1 \bar{\mathbf{x}}_1(k) + B_1 u(k) \quad (69)$$

$$e_{v,1}(k) = C_1 \bar{\mathbf{x}}_1(k) \quad (70)$$

with,

$$\begin{aligned} A_1 &= \begin{bmatrix} a_p & b_p C_{\bar{\xi}_1} \\ 0 & A_{\bar{\xi}_1} \end{bmatrix} = \begin{bmatrix} 0.9942 & 0.00528 \\ 0 & 1 \end{bmatrix}, \\ B_1 &= \begin{bmatrix} b_p \\ 0 \end{bmatrix} = \begin{bmatrix} 0.00528 \\ 0 \end{bmatrix}, \\ C_1 &= \begin{bmatrix} c_p & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \\ \bar{\mathbf{x}}_1(k) &= \begin{bmatrix} \bar{x}_p(k) \\ \bar{\xi}_1(k) \end{bmatrix}. \end{aligned} \quad (71)$$

Thus, based on the augmented system, the observer for stage 1 is formulated as:

$$\hat{\bar{\mathbf{x}}}_1(k+1) = A_1 \hat{\bar{\mathbf{x}}}_1(k) + B_1 u(k) + L_1 (e_y(k) - \hat{e}_{y,1}(k)) \quad (72)$$

$$\hat{e}_{y,1}(k) = C_1 \hat{\bar{\mathbf{x}}}_1(k) \quad (73)$$

where L_1 is designed such that the eigenvalues of the matrix $(A_1 - L_1 C_1)$ are located at $[0.9, 0.8]$, and the disturbance estimation can be retrieved from $\hat{\xi}_1(k) = \begin{bmatrix} 0 & 1 \end{bmatrix} \hat{\bar{\mathbf{x}}}_1(k)$.

7.3.2. Observer 2

Consider the internal model of an harmonic disturbance $\bar{\xi}_2(k)$, that emulates an harmonic interference on the load, with frequency $\omega_x = 17$ rad/s represented in state-space as:

$$\bar{\mathbf{x}}_{\bar{\xi}_2}(k+1) = A_{\bar{\xi}_2} \bar{\mathbf{x}}_{\bar{\xi}_2}(k), \quad \bar{\xi}_2(k) = C_{\bar{\xi}_2} \bar{\mathbf{x}}_{\bar{\xi}_2}(k) \quad (74)$$

with,

$$A_{\bar{\xi}_2} = \begin{bmatrix} 0 & 1 \\ -1 & 2 \cos(\omega_x T_s) \end{bmatrix}, C_{\bar{\xi}_2} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \bar{\mathbf{x}}_{\bar{\xi}_2}(k) = \begin{bmatrix} \bar{\xi}_2(k) \\ \bar{\xi}_2(k+1) \end{bmatrix}.$$

Then, the augmented system for the stage 2 is given by:

$$\bar{\mathbf{x}}_2(k+1) = A_2 \bar{\mathbf{x}}_2(k) + B_2 u(k), \quad e_{y,2}(k) = C_2 \bar{\mathbf{x}}_2(k) \quad (75)$$

with,

$$A_2 = \begin{bmatrix} a_p & b_p C_{\bar{\xi}_2} \\ 0_{2 \times 1} & A_{\bar{\xi}_2} \end{bmatrix} = \begin{bmatrix} 0.9942 & 0.00528 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 1.9928 \end{bmatrix},$$

$$B_2 = \begin{bmatrix} b_p \\ 0_{2 \times 1} \end{bmatrix} = \begin{bmatrix} 0.00528 \\ 0 \\ 0 \end{bmatrix}, C_2 = \begin{bmatrix} c_p & 0_{1 \times 2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix},$$

$$\bar{\mathbf{x}}_2(k) = \begin{bmatrix} \bar{x}_p(k) \\ \bar{\xi}_2(k) \\ \bar{\xi}_2(k+1) \end{bmatrix}.$$

Thus, based on the augmented system, the observer for stage 2 is formulated as:

$$\hat{\bar{\mathbf{x}}}_2(k+1) = A_2 \hat{\bar{\mathbf{x}}}_2(k) + B_2 u(k) + L_2 (e_y(k) - \hat{e}_{y,2}(k)) \quad (76)$$

$$\hat{e}_{y,2}(k) = C_2 \hat{\bar{\mathbf{x}}}_2(k) \quad (77)$$

where L_2 is designed such that the eigenvalues of the matrix $(A_2 - L_2 C_2)$ are located at $[0.98, 0.95, 0.9]$, and the disturbance estimation can be retrieved from $\hat{\xi}_2(k) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \hat{\bar{\mathbf{x}}}_2(k)$.

7.3.3. Observer 3

In the case of periodic signals, this disturbance could emulate the interference of the motor shaft with an external object, such that, at constant speed, the impact would represent a periodic signal. In order to develop the observer for periodic signals, the system Eqs.(64)-(65) is represented in transfer function form and taken as the nominal model for the observer design:

$$\bar{G}_p(z) = \frac{b_p}{z - a_p} = \frac{\hat{e}_y(z)}{u(z) + \hat{\xi}_3(z)} \quad (78)$$

Then, after some algebraic manipulations, the system is described as:

$$b_p u(z) + b_p \hat{\xi}_3(z) - e_y(z)(z - a_p) = \tilde{e}_{ey}(z)(z - a_p) \quad (79)$$

with $\tilde{e}_{ey}(z) = \hat{e}_y(z) - e_y(z)$. Therefore, by multiplying the term $(1 - z^{-N})$ to both sides of the equation, the system is defined as:

$$(1 - z^{-N}) b_p \hat{\xi}_3(z) = (1 - z^{-N})(z - a_p) \tilde{e}_{ey}(z) \quad (80)$$

where, the control law (periodic disturbance estimation) for the observer is formulated:

$$\hat{\xi}_3(z) = \frac{1}{(1 - z^{-N}) b_p} \left[-z^{-N}(z - a_p) \tilde{e}_{ey}(z) - \left(\frac{k_1 z + k_0}{z - 1} \right) \tilde{e}_{ey}(z) \right] \quad (81)$$

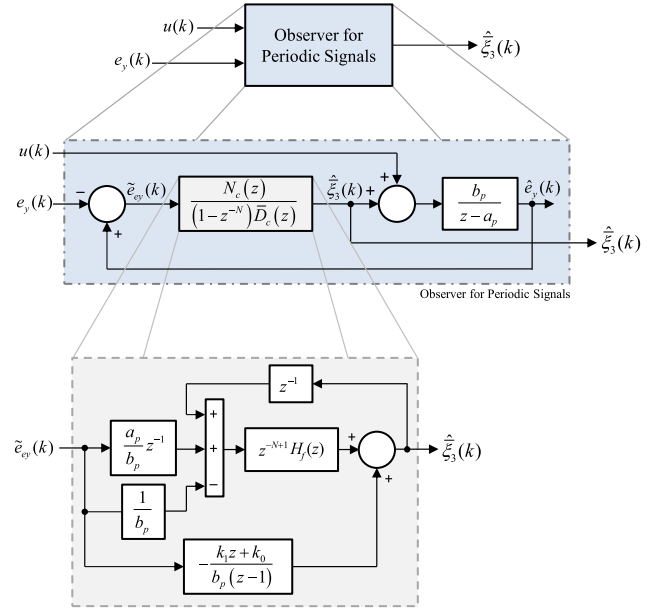


Fig. 7. Detailed implementation scheme of the observer for periodic signals.

with $N = \frac{T_p}{T_s} = 50$, and the fundamental period of the periodic signal $T_p = \frac{1}{4}$ sec. For implementation purposes (see Remark 4), this periodic estimator is filtered and expanded as follows:

$$\hat{\xi}_3(z) = z^{-N} H_f(z) \hat{\xi}_3(z) - z^{-N} \frac{(z - a_p)}{b_p} H_f(z) \tilde{e}_{ey}(z) - \frac{1}{b_p} \left(\frac{k_1 z + k_0}{z - 1} \right) \tilde{e}_{ey}(z), \quad (82)$$

and implemented as shown in Fig. 7, where

$$H_f(z) = \frac{h_5 z^{10} + h_4 z^9 + \dots + h_1 z^6 + h_0 z^5 + h_1 z^4 + \dots + h_4 z + h_5}{z^5}. \quad (83)$$

Notice that $H_f(z)$ is a 10th-order zero-phase FIR filter with the following parameters: $h_5 = 0.00506$, $h_4 = -0.01264$, $h_3 = 0.03398$, $h_2 = -0.06350$, $h_1 = 0.08928$, $h_0 = 0.89561$ and it was obtained with the following Matlab command `fir1(10,0.9)`.

The control gains k_0 and k_1 of Eq.(81) are designed finding the closed-loop system for this observer, thus replacing the control law Eq.(81) into the system Eq.(80) the following dynamics for $\tilde{e}_{ey}(z)$ is derived:

$$-z^{-N} \tilde{e}_{ey}(z) - \frac{k_c}{z - 1} \tilde{e}_{ey}(z) = (1 - z^{-N}) \tilde{e}_{ey}(z) \quad (84)$$

$$(z - 1 + k_c) \tilde{e}_{ey}(z) = 0 \quad (85)$$

where the controller parameters are easily determined as following: $k_1 = 1 - p_{des} = 0.02$, $k_0 = -a_p k_1 = -0.0198$ where $k_1 = k_c$ and $p_{des} = 0.98$ is the desired dominant dynamics of $\tilde{e}_{ey}(z)$.

Please note that this procedure resembles a control scheme; however, it is actually a method for estimating the periodic disturbance $\hat{\xi}_3(k)$ by utilizing the duality of control and observer strategies.

7.3.4. Control law

Given the transfer function of the system: $\bar{G}_p(z) = \frac{b_p}{z - a_p} = \frac{e_y(z)}{u_L(z)}$, then, a proportional stabilizing control law is designed: $u_L(z) = -k_s e_y(z)$, where the tracking error dynamics $e_y(z)$ is dominated by the following polynomial: $z - a_p + k_s b_p = 0$, and the stabilizing control gain k_s is designed as: $k_s = \frac{a_p - p_{ds}}{b_p} = 0.1403$ where p_{ds} is the desired tracking error dynamics.

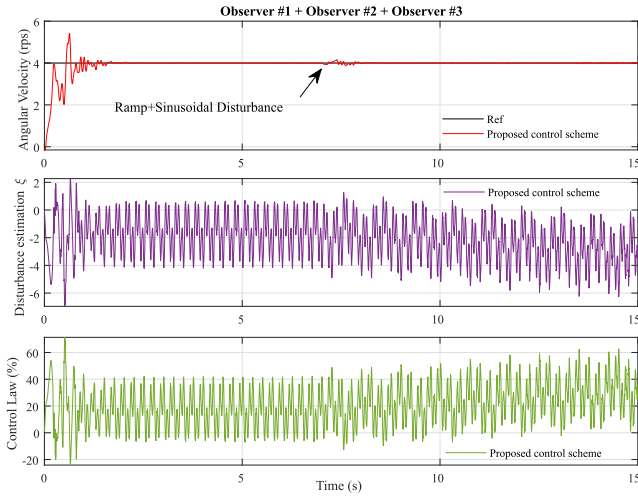


Fig. 8. Experimental results of the proposed control scheme using all Observer Stages for periodic disturbance rejection with ramp + sinusoidal load torque at 7 sec.

Table 1

Performance indices comparison of the observers across the three studied cases.

Case	Index	ESO	GESO	HGO	Proposed
Case 1	ISE	0.0626	0.0675	0.1227	0.1074
	ITAE	6.1694	6.2107	8.5660	2.4106
	Control Energy	2022.27	2156.98	2014.72	2080.48
Case 2	ISE	0.6139	unstable	0.9776	0.1959
	ITAE	18.1043	unstable	23.3591	2.4734
	Control Energy	5069.40	unstable	4409.22	5240.22
Case 3	ISE	1.5158	unstable	2.2418	0.2273
	ITAE	29.8345	unstable	35.9797	2.1556
	Control Energy	9517.66	unstable	7206.77	9683.12

7.4. Experimental results and analysis

This section presents the experimental results of the proposed control scheme using the testbed described above. The objective is to control the velocity of the main motor, denoted as $y(k)$, either by tracking or rejecting a periodic signal in the presence of torque-load disturbances generated by the secondary motor.

7.4.1. Periodic disturbance rejection

The first experiment was conducted to regulate the motor velocity at 4 rps and to reject periodic disturbances generated by a magnetic setup, combined with an external ramp and sinusoidal load torques. In this test, Observers 1, 2, and 3 were enabled and utilized along with the control law. Fig. 8 demonstrates successful rejection of all disturbances. The experimental results show an effective rejection of the periodic disturbance including the external load torques. Note that at 7 sec the external load torque is applied, and then the control system modifies its control law to provide the disturbance rejection according to the applied disturbance so that the control law and the estimation of the disturbance ξ combine all disturbance components.

7.4.2. Periodic signal tracking

In order to evaluate periodic signal tracking, the following reference signal is defined:

$$y^*(k) = 4 + 0.3 \sin(8\pi kT_s) + 0.3 \sin(16\pi kT_s) + 0.4 \sin(24\pi kT_s), \quad (86)$$

A tracking test was performed using all observer stages. In this experiment, the load disturbance applied at 7 seconds consisted of a combination of a ramp and a sinusoidal signal. Fig. 9 presents the experimental results, which demonstrate successful tracking of the periodic

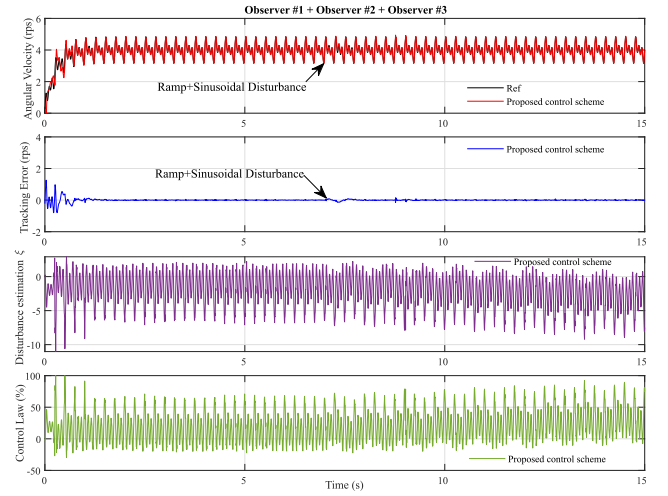


Fig. 9. Experimental results of the proposed control scheme using all Observer Stages for tracking a periodic signal while rejecting a ramp disturbance combined with a sinusoidal disturbance.

reference signal and effective rejection of the overall disturbance, as evidenced by the tracking error. These results also confirm that the tracking and disturbance rejection capabilities are preserved, along with a robust transient response under these conditions. The figure additionally shows the estimation of the disturbance signal, ξ , provided by the proposed scheme. This estimation exhibits periodic components primarily due to the reference signal and changes when the disturbance is introduced, ensuring adequate tracking and rejection within the control scheme.

7.4.3. Comparison with other observers

This section shows an extensive set of experimental tests conducted to provide a clearer and more comprehensive performance comparison of the proposed scheme against well-known disturbance observers such as: Extended-state Observer (ESO), Generalized Extended-state Observer (GESO), and High Gain Observer (HGO). These experimental results are organized into three distinct datasets:

Case 1: The main objective is speed regulation and disturbance rejection when the system is subjected to a combination of a step disturbance and an inter-harmonic disturbance.

Case 2: The primary goal is speed regulation and rejection of a periodic disturbance combined with step and an inter-harmonic disturbance.

Case 3: The fundamental objective is periodic reference tracking under the simultaneous injection of a step disturbance and an inter-harmonic disturbance.

Fig. 10 presents the experimental results of Case 1. From 0-7 s, the system operates without disturbances (apart from modeling uncertainties and minor unbalances). From 7-15 s, a load torque disturbance is applied, consisting of a step combined with an inter-harmonic component at 17 rad/s. It is worth noting that different observers (medium bandwidth tuned) are evaluated while keeping the same control structure. When the combined disturbance is injected, all controllers react to counteract it (as evidenced by the offset and harmonic components in the control law). However, the proposed scheme is the only one capable of almost completely rejecting the disturbance in steady state. Additionally, three performance indices were computed (see Table 1): ISE, ITAE, and control energy. Regarding the ISE index, the proposed scheme is penalized during the transient response due to a slight overshoot caused by the observers' relatively low bandwidth. In contrast, the proposed approach achieves the best ITAE values, thanks to its superior steady-state

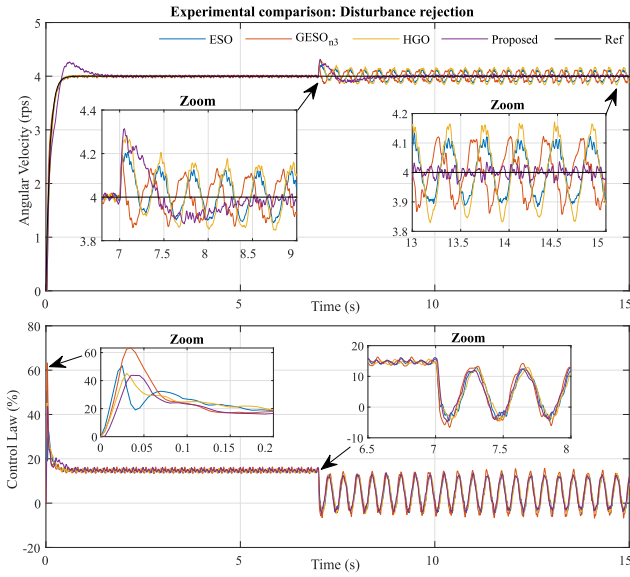


Fig. 10. Comparison of Experimental Results: Case 1.

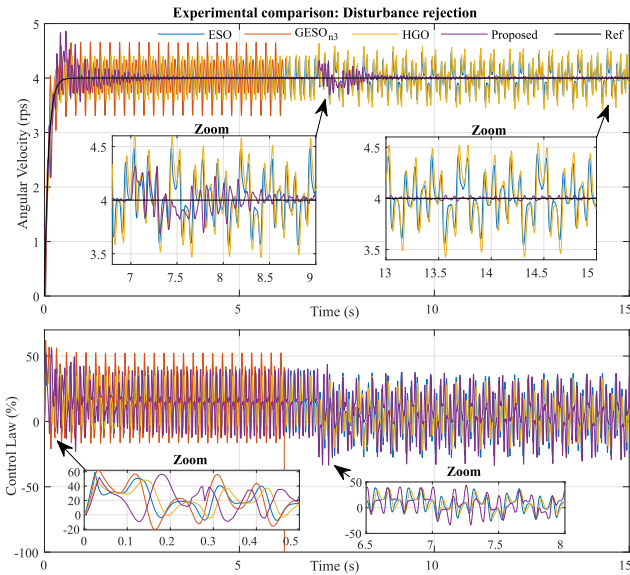


Fig. 11. Comparison of Experimental Results: Case 2.

performance. Finally, in terms of control energy, the proposed scheme delivers significantly improved disturbance rejection with only an approximately $\pm 3\%$ variation in energy consumption.

Fig. 11 shows the experimental results for Case 2. A periodic disturbance with a fundamental frequency of 4 Hz is applied throughout the entire experiment (0–15 s). At $t = 7$ s, an additional load torque disturbance is injected, consisting of a step combined with an inter-harmonic component at 17 rad/s. The proposed scheme is the only one capable of almost completely rejecting all disturbances in steady state (see zoom between 13–15 s). Moreover, its transient response is comparable to the other schemes (see zoom between 7–9 s). Furthermore, the initial peak of the control law of the proposed scheme (see zoom between 0–0.5 s) remains lower than that of the compared approaches, preventing actuator saturation thanks to its low observer bandwidth. The proposed approach yields the best ISE and ITAE values (see Table 1) across this dataset, owing to its superior steady-state and post-transient performance. In terms of control energy, the proposed scheme achieves significantly enhanced disturbance rejection with an increase of energy consumption of 15.8%

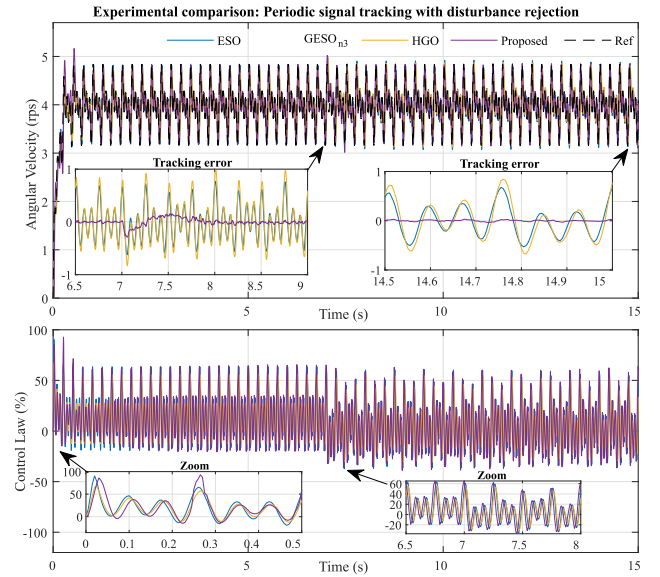


Fig. 12. Comparison of Experimental Results: Case 3.

relative to HGO and only a modest increase of approximately 3.2% relative to ESO.

Fig. 12 shows the experimental results for Case 3. A periodic reference (as defined in Eq.(86)) for tracking is applied throughout the entire experiment (0–15 s). At $t = 7$ s, an additional load torque disturbance is injected (a step combined with an inter-harmonic of 17 rad/s). The figure shows that the proposed scheme is the only one capable of almost completely track the periodic reference while rejecting all disturbances (see tracking error plots detailing zoom between 6.5–9 s and 14.5–15 s). Moreover, its transient response at 7 sec is better than that of the other schemes (see tracking error zoom between 6.5–9 s). Furthermore, the initial peak of the control law of the proposed scheme (see zoom between 0–0.5 s) remains comparable than the other approaches. The proposed approach achieves the lowest ISE and ITAE values (see Table 1). However, in terms of control energy, the proposed scheme achieves substantially improved disturbance rejection at the cost of a 25% increase in energy consumption relative to HGO, and only a modest 1.7% increase relative to ESO. This additional control effort is largely justified by the significant reduction in tracking error.

To evaluate the peak response and robustness of the HGO against the proposed scheme, extensive experimental tests were performed. Fig. 13 compares both approaches in detail. As ϵ decreases, HGO performance indices improve, but at the cost of increased system vibrations and occasional instability. Under a complex periodic disturbance, the proposed scheme (medium-bandwidth tuned) maintains reasonable control energy, limits the maximum control action to about 73%, and achieves superior disturbance rejection. In contrast, the HGO exhibits poor steady-state rejection, as highlighted in the zoomed region between 13 and 15 s. Conventional observers like ESO and HGO require high bandwidth to estimate periodic disturbances, which can compromise stability due to noise sensitivity and peaking. To address this, advanced designs use IMP-based extended states.

Alternative approaches for handling time-varying disturbances include the Robust Integral of the Sign of the Error (RISE) control strategy (Patre et al., 2008; Xian et al., 2004), which achieves asymptotic (or semi-global asymptotic) tracking without requiring explicit disturbance models. However, standard RISE designs often rely on high feedback gains to dominate disturbance bounds and their derivatives, potentially leading to conservative performance, amplified measurement noise, peaking phenomena, and increased control effort (Wen et al., 2021; Yang et al., 2022). In contrast, the proposed scheme, incorporating states derived from the Internal Model Principle, enables structured

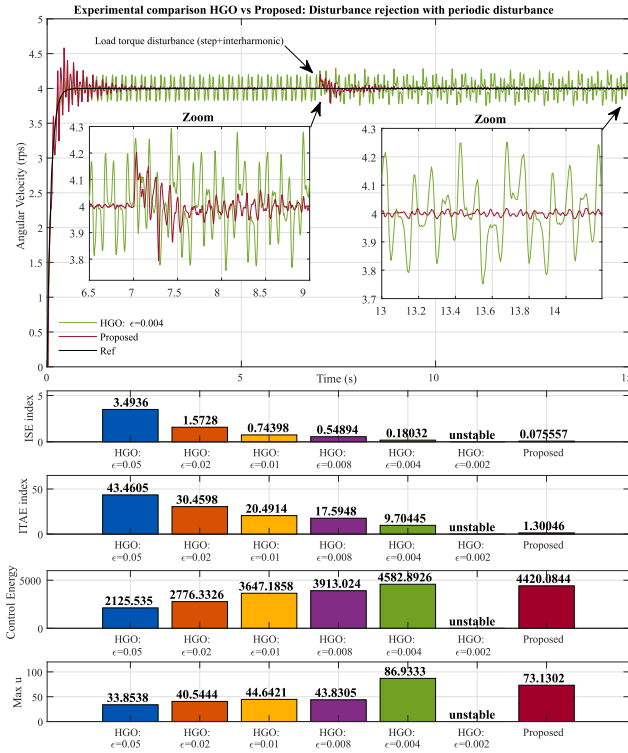


Fig. 13. Experimental results for comparison with HGO.

estimation and cancellation of periodic or inter-harmonic disturbances, resulting in reduced observer bandwidth, lower gain requirements, and enhanced robustness to noise.

8. Conclusions

This work presents a novel observer-based control scheme designed for high-performance disturbance rejection. The proposed approach introduces a fully multi-decoupled observer scheme based on the Internal Model Principle (IMP), where for each non-periodic disturbance, an individual observer employs an Extended State Observer (ESO) with the disturbance's internal model as extended states. This scheme addresses several challenges associated with classical methods, including precise disturbance rejection, as ensured by the Internal Model Principle, and the issue of large observer dimensions. By reducing the dimension of each observer compared to a single ESO observer, the design process becomes more straightforward. This feature is a critical novelty as it aims to execute a high-performance disturbance rejection task without increasing the numerical or dimension complexity of the scheme. Additionally, the scheme effectively handles multiple disturbances by allowing for the exact rejection of new, well-known disturbances without the need for a complete redesign—simply by adding a new parallel observer. The approach also avoids the high-gain issues that often arise with classical observers of similar rejection capability due to their larger dimensions.

The proposed scheme offers further advantages, such as the robustness to selectively activate or deactivate observers without compromising stability, and a significant reduction in computational cost during experimental implementation compared to a classical single ESO approach. Specifically, it reduces the number of mathematical operations required for implementation on a System-on-Chip (SoC) device. While the proposal does not aim to outperform classical methods in terms of rejection capability—since any scheme using the same number of internal models should achieve comparable performance—it does offer significant improvements in both design efficiency and structural flexibility. The proposed scheme features the ability to implement multiple

lower-order observers to address high-order polynomial disturbances, effectively achieving the desired polynomial order. This approach simplifies the design process and enhances the management of high-gain observers, which are typically challenging in classical methods. Additionally, the scheme is designed to track any type of reference signal, leveraging an error-based framework where the reference is treated as an additional disturbance to the system.

Additionally, the proposal introduces a novel observer structure for managing general periodic disturbances without resorting to large-dimension observers. This periodic observer integrates seamlessly with the decoupled observer scheme, enabling comprehensive disturbance rejection in practical applications.

The effectiveness of the proposed scheme is demonstrated through a mechanical experimental setup where various disturbances are applied to a DC motor via a secondary coupled DC motor. The system also includes a rotating magnet configuration to simulate periodic disturbances. Experimental results clearly demonstrate the exact rejection capability of the proposed scheme for the disturbances considered, while the detailed design process highlights the scheme's ease of use and advantages. Finally, the proposed scheme successfully achieves disturbance rejection through a simple, parallel, decoupled, and effective design.

CRedit authorship contribution statement

John Cortés-Romero: Conceptualization, Writing – original draft, Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Formal analysis; **Horacio Coral-Enriquez:** Writing – review & editing, Validation, Methodology, Formal analysis; **Brian Camilo Gómez-León:** Writing – review & editing, Formal analysis, Methodology, Investigation; **Hebertt Sira-Ramírez:** Supervision, Methodology, Investigation.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Periodic signals algebra

Let $\xi_{p1}(k)$ and $\xi_{p2}(k)$ N -periodic and T the sample time. Therefore: $\xi_{p1}(k) = \xi_{p1}(k - N)$, $\xi_{p2}(k) = \xi_{p2}(k - N)$.

1. The addition or subtraction of two periodic signals is also periodic. Let $\xi_{p3}(k) = \xi_{p1}(k) \pm \xi_{p2}(k)$. Then:

$$\begin{aligned} \xi_{p1}(k) \pm \xi_{p2}(k) &= \xi_{p1}(k - N) \pm \xi_{p2}(k - N) \\ \Rightarrow \xi_{p3}(k) &= \xi_{p3}(k - N) \end{aligned}$$

2. The multiplication or division of two periodic signals is also periodic. Let $\xi_{p4}(k) = \xi_{p1}(k) * \xi_{p2}(k)$. Then:

$$\begin{aligned} \xi_{p1}(k) * \xi_{p2}(k) &= \xi_{p1}(k - N) * \xi_{p2}(k - N) \\ \Rightarrow \xi_{p4}(k) &= \xi_{p4}(k - N) \end{aligned}$$

in the case of division, for a correct definition, it is necessary that ξ_{p2} does not cross zero.

3. The discrete-time derivative of a periodic signal is also periodic. Let $\xi_{p5}(k) = \frac{\xi_{p1}(k) - \xi_{p1}(k-1)}{T}$. Then:

$$\begin{aligned} \frac{\xi_{p1}(k) - \xi_{p1}(k-1)}{T} &= \frac{\xi_{p1}(k - N) - \xi_{p1}(k - 1 - N)}{T} \\ \Rightarrow \xi_{p5}(k) &= \xi_{p5}(k - N) \end{aligned}$$

4. Unlike the derivative with respect to k , the set of periodic signals does not remain closed with respect to integration. A simple counterexample is a constant signal, since its integral is a ramp which is not constant.

Steady-state response of periodic signals

On the other hand, we are interested in analyzing the effect that a filter has on a periodic signal. In particular, it will be analyzed for a stable filter with a Jury denominator polynomial. Periodic signals are known to be expressible as a summation of sine and cosine functions; i.e., if $\xi_p(k)$ is a periodic signal of period N , then we may write it as a sum of harmonics of the fundamental frequency $2\pi/(NT)$:

$$\xi_p(t) = a_0 + \sum_{i=1}^{\infty} \left[a_i \cos \frac{2i\pi}{N} k + b_i \sin \frac{2i\pi}{N} k \right]$$

A simple steady-state analysis can be derived from frequency response theory. Suppose that the signal $\xi_p(k)$ is processed by a stable transfer function $G_f(z)$ acting as a filter, therefore the corresponding steady-state response is given by

$$\xi_{ss}(k) = \bar{a}_0 + \sum_{i=1}^{\infty} \left[\bar{a}_i \cos \left(\frac{2i\pi}{N} k + \theta_i \right) + \bar{b}_i \sin \left(\frac{2i\pi}{N} k + \theta_i \right) \right]$$

where $\bar{a}_0 = a_0 G_f(0)$, $\bar{a}_i = a_i |G_f(j2i\pi/N)|$, $\bar{b}_i = b_i |G_f(j2i\pi/N)|$, $\theta_i = \angle G_f(j2i\pi/N)$, with $|G_f(j2i\pi/N)|$ and $\angle G_f(j2i\pi/N)$ respectively denoting the magnitude and phase of $G_f(j2i\pi/N)$.

It can be seen from the last two equations that the steady state response preserves the structure of the periodic input signal except for changes in magnitude and phase shifts. Therefore, it can be concluded that the steady-state response of the filter output is also a periodic signal of period N .

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